

Aayam Regmi

Comparative Evaluation of Vision-Based Finger Tracking for Piano Interaction

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Production Project

Comparative Evaluation of Vision Based Finger Tracking for Piano Interaction

Acknowledgement

I would like to thank my supervisor, Sir Saroj Sharma, and my module leader, Dr Rohit Raj Pandey, for providing me with feedback and academic guidance throughout the lifecycle of this research. The regular meetings and feedback from supervisors have been an invaluable experience and have helped in refining the methodology and acting according to the project schedule. I would also extend my gratitude to the participants who were willing to volunteer in dataset collection, stepping into an unfamiliar domain of musical instruments and artificial intelligence.

Abstract

Modern technologies like virtual reality, human-computer interaction, and tracking software use vision-based models. For music and instruments applications, vision-based hand tracking models face the obstacle of rapid finger movements and overlaps when tracking hands playing any piano performance. Systematic benchmarks for playing a piano under realistic domestic conditions need to be reviewed. This research evaluates two widely used open-source hand tracking models: MediaPipe and OpenPose. Both models are tasked to identify which finger strikes which key during piano performance. A custom dataset collection pipeline was developed to capture video data positioned overhead and MIDI signals as ground truth from a CTK-2400 synchronized together. The first sessions had base variables set that were Fitzpatrick skin type, ambient illuminance, and hand size to test for performance across participant demographics. A novel horizontal axis **Mean Per Joint Position Error (MPJPE) metric** was calculated to account for finger length and depth artefacts, across 38 sessions for each identified key press, to evaluate the performance after finding the detection rate. Also, a polygon box check system using a homography calibration matrix was developed to capture if landmark were within the premise of the key's surface area. Note events 4777 were collected, MediaPipe achieved 6.07 px mean MPJPE with a polygon hit rate of 63.3%, while OpenPose recorded 5.87 px MPJPE but on only 8.1% of events due to its requirement for pre-cropped hand input. Skin tone, hand size, and lighting were tested, and no significant errors were caused by these metrics when using MediaPipe. A substantial asymmetry for left- and right-hand accuracy was observed in MediaPipe's polygon hit rate of left 53.2% and right 95.8%, even if the video data was taken in the same octave and same keys for both hands, identifying a previously undocumented camera-geometry interaction. The dataset contributes as having an MPJPE metric along with timestamps of MIDI and video data, and homography calibrated polygon mask methodology as an open-source contribution able to be used on any new vision models for benchmarking and other vision-based piano interaction results.

Student Declaration

I, Aayam Regmi (Student ID: 77466864), declare that this report is my own original work submitted with fulfilled requirements, ethical consent of Leeds Beckett University for the production project of BSc (Hons) Computing and The British College. All cited references are in accordance with the Harvard referencing format.

Table of Contents

Acknowledgement.....	3
Abstract	4
Student Declaration.....	5
Table of Contents	6
List of Resources	10
List of Figure	10
List of Tables	11
List of Abbreviations.....	11
1. Introduction	13
2. Review of literature	14
3. Review of technologies	17
3.1 Major Hardware Components.....	17
3.2 Software Components and Libraries.....	19
4. Methodology	20
4.1 Methodology description	20
4.2 Research design plan	20
4.3 Methodology defense.....	21
5. Research/ Product design	21
5.1 Data collection pipeline system	22
6. Implementation and Testing.....	24
6.1 Dataset Collection Pipeline Overview	24
6.2 Participant and Environmental Characterization	24
6.2.1 Skin Tone Classification (Fitzpatrick Scale)	24
6.2.2 Ambient Illuminance Classification.....	25
6.2.3 Hand Size Classification	26
6.3 Pipeline MPJPE Calibration and Bias Reduction.....	26
6.3.1 Horizontal MPJPE: Fingertip-to-Key Centre Offset	27
6.3.2 Polygon Containment for Finger Detection	27

6.3.3 Perspective Homography Calibration	29
6.4 Synchronized Multi-modal Data Acquisition	32
6.4.1 MIDI–Video Temporal Alignment	32
6.5 System Performance and Detection Statistics	32
6.6 Participant Study Design	34
6.6.1 Participant Recruitment and Inclusion Criteria	34
6.6.2 Session Protocol and Procedure	35
6.6.3 Dataset Structure and Storage	35
6.7 Offline MPJPE Analysis	35
7. Product Evaluation	36
7.1 Functional Requirements Completion	36
7.2 Non-Functional Requirements Completion	38
7.3 Potential Benchmark Application	39
8. Research Evaluation	39
8.1 Overall Model Performance	40
8.1.1 Detection Rate	40
8.1.2 Hand Assignment and Per-Finger Computation	41
8.1.3 Per-Hand and finger detection and match rate	42
8.1.4 Detection-Fail Rate by Fitzpatrick Type	44
8.1.5 Detection-Fail Rate by Lighting Condition	45
8.1.6 Detection Outcome Breakdown	45
8.2 MPJPE outcomes	46
8.2.1 MPJPE	46
8.2.2 2D Mask Perspective Note	47
8.2.3 MediaPipe Per-Finger MPJPE and Polygon Hitrate	48
8.2.4 OpenPose Per-Finger MPJPE	49
8.2.5 MPJPE per finger Outcome Breakdown	50
8.3 Effect of Skin Tone (Fitzpatrick Type) on MPJPE	51
8.3.1 Computation of skin tone type	51

8.3.2 Fitzpatrick Type and MPJPE	51
8.3.3 MPJPE by Fitzpatrick Breakdown	52
8.4 Effect of Ambient Illuminance on MPJPE	53
8.4.1 Computation of lighting	53
8.4.2 MPJPE by Lighting Condition.....	53
8.4.3 MPJPE by Lighting Breakdown	55
8.5 Effect of Hand Size on MPJPE	55
8.5.1 Computation of Hand Size	55
8.5.2 MPJPE vs Hand Size.....	56
8.5.3 Hand size effect breakdown	56
9. Summary and Conclusion	57
9.1 Summary of Project Aim and Outcomes.....	57
9.2 Novel Contributions	58
9.3 Room for Future Development	59
10. Bibliography.....	60
11. Appendices	63
A. Initial Plan Document.....	63
B. Risk Register Document	72
C. Ethics Approval	74
D. Participant Consent Form	88
E. Source Code Snippets.....	88
config.py	88
camera_setup.py	88
lux_calculator.py	89
fitzpatrick_detector.py.....	89
midi_recorder.py	89
model_manager.py.....	90
stats_collector.py	91
key_calibration.py	91

record.py.....	91
live_preview.py.....	92
recalibrate.py	93
analyse.py	93
plot_results.py	94
export_table_data.py.....	94
data_summary.py	95
F. Supervisor Meeting Records.....	Error! Bookmark not defined.

List of Resources

List of Figure

- Fig 3.1.a: CTK-2400 electrical piano
- Fig 3.1.b: USB 2.0 to USB 3.0 cable
- Fig 3.1.c: Google Pixel 7 and USB Type-C cable
- Fig 3.1.d: Velbon adjustable camera tripod
- Fig 3.1.e: MSI gaming laptop
- Fig 4.2.a: project timeline and Gantt chart
- Fig 5.1.a: Data collection pipeline setup
- Fig 5.1.b: Session setup parameters
- Fig 5.1.c: System Architecture Diagram
- Fig 5.1.d: Use Case Diagram
- Fig 6.2.1.a: FitzPatrick scale (FitzPatrick, 1988)
- Fig 6.2.2.a: LUX Meter Mobile Phone (14.67 lux)
- Fig 6.3.1.a: MediaPipe hands landmark preview
- Fig 6.3.2.b: polygon containment mask
- Fig 6.3.2.c: polygon containment mask preview
- Fig 6.3.2.d: Polygon containment mask controls
- Fig 6.3.3.a: homography calibration mask
- Fig 6.3.3.b: correct landmark detection
- Fig 6.3.3.c: out of polygon mask false detection
- Fig 6.3.3.d: mask visualization after accounting for vertical finger depth
- Fig 6.5.a: live_preview statistics
- Fig 6.5.b: live_preview.py per hand per finger result
- Fig 6.3.3.a: Raw data saved structure
- Fig 6.7.a: Offline processing UI (analyse.py -ui)
- Fig 7.3.a: Prototype UI design benchmark app
- Fig 8.a: test data diversity charts
- Fig 8.1.1.a: MediaPipe vs OpenPose detection rate chart
- Fig 8.1.3.a: Per Hand and finger detection rate for MediaPipe
- Fig 8.1.3.c: Average Finger detection rate for OpenPose (Low n detection rate)
- Fig 8.1.4.a: detection fail rate by skin tone
- Fig 8.1.5.a: detection fail rate by lighting condition
- Fig 8.2.1.a: Overall MPJPE on MediaPipe vs OpenPose
- Fig 8.2.2.a: Black keys 2D calibration mask misalignment
- Fig 8.2.3.b: finger breakdown for MPJPE (MediaPipe)
- Fig 8.2.3.c: finger breakdown for MPJPE (MediaPipe)
- Fig 8.2.3.d: MediaPipe per hand polygon hit rate

- Fig 8.2.4.b: Overall MPJPE combined for both hands (OpenPose)
- Fig 8.3.2.b: Average MPJPE per Fitzpatrick scale type (II-IV)
- Fig 8.4.2.b: Mean MPJPE per lighting category (no bright conditions were recorded)
- Fig 8.4.2.c: line graph for MPJPE vs Lux levels
- Fig 8.5.2.a: Hand size and MPJPE correlation chart

List of Tables

- Table 7.1.a: Function Requirements Summary
- Table 7.2.a: Nonfunctional summary
- Table 8.1.1a: Model Summaries
- Table 8.1.3.b: Per finger match rate MediaPipe
- Table 8.1.3.d: Average Finger detection rate for OpenPose Summary
- Table 8.1.4.b: detection fail rate by skin tone
- Table 8.1.5.b: detection fail rate by lighting condition summary
- Table 8.2.1.b: Overall MPJPE on MediaPipe vs OpenPose Summary
- Table 8.2.3.a: Per finger MPJPE Summary
- Table 8.2.4.a: OpenPose per finger summary
- Table 8.3.2.a: Fitzpatrick type MPJPE summary
- Table 8.4.2.a: MPJPE by lighting condition Summary
- Table 8.4.2.d: MPJPE vs Lux Levels Summary
- Table 8.5.2.b: Hand size and MPJPE correlation Summary

List of Abbreviations

- API – Application Programming Interface
- BGR – Blue, Green, Red
- CIELab – Commission Internationale de l'Éclairage (Lab colour space)
- CLI – Command-Line Interface
- CNN – Convolutional Neural Network
- CSV – Comma-Separated Values
- CV – Computer Vision
- DR – Detection Rate
- FPS – Frames Per Second
- GPU – Graphics Processing Unit
- GUI – Graphical User Interface
- HCI – Human-Computer Interaction
- ITA – Individual Typology Angle
- JSON – JavaScript Object Notation
- MIDI – Musical Instrument Digital Interface

- MoSCoW – Must, Should, Could, Won't (prioritisation method)
- MPJPE – Mean Per-Joint Position Error
- PAF – Part Affinity Field
- RAD – Rapid Application Development
- RGB – Red, Green, Blue
- ROI – Region of Interest

1. Introduction

This research aims to evaluate various metrics of widely used vision-based hand tracking models. MediaPipe hands and OpenPose are some frameworks that will be mainly focused on. The ground truth to evaluate the metrics will be an electric piano (CTK-2400). The captured MIDI data synchronized with video footage timestamps will ensure a fair comparison between both models. An overhead camera will display finger landmarks and the position of these landmarks. When a MIDI signal is detected, the calculation of the distance between the played key and the landmarks will be computed. According to a certain threshold value, the calculation will be deemed accurate or false flags. The research also aims to show if any correlations exist between hand tracking models, lighting conditions, hand size and skin tone diversity. Skin tone diversity is considered as a metric here to evaluate if there is any disparity between diverse groups in context of hand tracking. Buolamwini and Gebru (2018) assessed vision systems in context of various skin tone groups across various skin tone groups demonstrating accuracy disparities. This became an evaluation variable for fixing diversity bias. Mehrabi et al. (2021) described that dataset homogeneity while training a model is a primary source of bias. The results would provide clear proposals for potential improvements on computer vision models backed by controlled data collection methods and give recommendations for training music-related models.

'Comparative Evaluation of Vision-Based Finger Tracking for Piano Interaction' combines several fields of applied computing. Real-time pose estimations, visual landmarks, temporal tracking, and HCI are some blends of this topic. Algorithm fairness is also a big discussion, as training a vision model requires a fair dataset with fairness to all groups of people. The piano was selected as ground truth for this research due to its unique design and finger tracking potential. Playing a piano piece can be rigorous for hand tracking models, especially when going from slow to fast rhythm and finger overlaps. The piano design shows closely packed keys, which could produce localization errors as well as temporal smoothness issues. These factors need to be considered when using a single camera vision tracking model that relies on frame RGB analysis. There are three main variables mentioned before to be considered while stress testing vision models. The first variable is testing multiple levels of illuminance. The SI unit of illuminance is lux, which measures the amount of visible light on the surface. The second variable is participants hand size, from one end to the other of the palm, measured manually. This variable was considered to test any localization errors between various hand sizes. The third variable is skin tone for algorithmic fairness. The Fitzpatrick scale (Fitzpatrick, 1988) will be used to categorize participants' skin tone data for later analysis. The Fitzpatrick scale was developed and published in 1988, which classifies skin pigmentation and reaction to UV light under 6 types (I-VI). Cho et al. (2023) stated that the Fitzpatrick phototype provides a fair scale for evaluations rather than subjective labelling.

The remainder of the report follows a structure that will explore the existing literature in this related work of vision-based hand tracking, single-camera RGB systems, and prior piano-related projects. Chapter 3 reviews the required hardware and software technologies used in the research data pipeline, which includes the CTK-2400 keyboard, overhead camera setups, the open-source models, and other Python libraries. Methodologies of development and design are also discussed using Rapid Application Development and the defense argument of why RAD was used. The following chapters present the product development, how each parameter is taken and tested to make sure the collection pipelines are working properly, about the polygon containment mask to detect landmark failures, and even video and MIDI data synchronization systems. The next chapters summarize what was found on the data using matplotlib graphs and tables, and break down the outcome of MPJPE, Fitzpatrick skin type, ambient illuminance, and hand size. The end of the report summarizes the outcomes, the novel contributions of the research, and the recommendations for improvements for similar future work.

2. Review of literature

The Xbox 360's Kinect, developed by Microsoft, was one of the earliest consumer-grade hardware finger tracking systems. It featured pose estimations for video games as well. The main restriction the Kinect posed was requiring depth-sensing hardware, imposing a three-dimensional constraint. The fine-tuned conditions required for tuning such hardware meant an embedded system developer had to consider an increase in hardware cost and code complexity. Han et al. (2013) reviewed the Kinect in hand gesture analysis and achieved great performance while considering the hardware constraints and complexity imposed by the device. Modern frameworks utilizing deep learning-based, RGB camera tracking systems have drastically changed the landscape of vision systems. Training convolutional neural networks (CNN) for classification jobs is the go-to method for vision systems today. Well-annotated hand images, diverse lighting conditions, and skin tone diversity meant highly accurate tracking results. With even a single camera system, joint localization is possible in real time. Experiments with low hardware cost are now possible in jobs such as vehicle number plate analysis, VR, and HCI systems.

A standardized approach for hand anatomy was introduced due to the transition from depth-based systems to single camera real time RGB landmark detection system. The approach followed by most landmark-based models is outputting a fixed skeleton of joint coordinates per frame, rather than processing all the pixels for each frame. This provides a lightweight and consistent downstream system with structured positional data. Molchanov et al. (2016) demonstrated that such a landmark identification system can achieve reliable classification without specialized depth sensor hardware. The ability to run such models on standard consumer-grade hardware, like a CPU, meant an expanded usability. This accessibility lowered a barrier for developers to build certain applications

like sign language recognition tools, gesture-based interfaces, and HCI systems. An everyday tutor and feedback system based on musical instruments is a strong candidate for research using the most relevant models to evaluate their suitability in rapid finger movement musical performances. Mueller et al. (2018) demonstrated a joint coordinate RGB-based system and approach accuracy like systems with depth sensor hardware in controlled environments.

MediaPipe Hands, developed by Zhang et al. (2020) at Google Research, is the most relevant open-source model for this research. The principle that this framework works on is identifying palm borders by hand-bounding boxes. The landmark model features 21 three-dimensional coordinates, with a detected bounding region. These landmarks translate into biological parts and joints of the human hand, including the fingertip. MediaPipe Hands is notable for being a lightweight model. Performance of 30 frames per second on a CPU alone is showcased by this model. This makes this framework suitable for real-time vision-related tasks. However, a common reported problem from the open-source community is that occlusion poses a significant performance loss. In the case of the piano, occlusion occurs when fingers overlap in a music performance, making it a common performance loss factor.

The OpenPose framework represents a different approach for finger tracking. Developed by Cao et al. (2019) at Carnegie Mellon University, it uses two-dimensional vector fields to encode joint

direction and relation across the image, establishing spatial relationships. This approach can detect and associate multiple human features, including hand key points, simultaneously through the network via a single forward pass. Jo and Kim (2022) concluded in their comprehensive study that, PoseNet, and Movenet concluded that at the cost of computing power, OpenPose achieved higher accuracy than other lighter models in multi-person scenarios. Chung et al. (2022) studied and concluded that, at the price of time and performance, heavier architecture consistently outperformed light models in terms of accuracy. Notable limitation of OpenPose is performance on CPU hardware falling below 15-20 FPS, and a lack of temporal smoothing features. In a piano setting the lack of temporal smoothing might cause keypress false flags, which directly entail accuracy reduction.

In the context of music, finger tracking systems have been explored in the instrument learning context. Liang et al. (2016) developed a virtual piano system with real-time hand interaction controls using an RGB-D camera. The fingertip tracking keypress features inverse kinematics and finger trajectory predictions. The system showed over 90% accuracy on finger press detection. A drawback is that the virtual piano requires specialized depth-sensing hardware in addition to the standard RGB input. This setup couldn't be further refined to a more consumer-friendly setup due to this. Hiranaka Grown-

Haeberli and Xue (2022) later used MediaPipe hands in the same domain of an augmented reality piano application for a purely RGB-based system. The augmented reality system achieved 81% accuracy at 12 frames per second. An improvement was that this system was designed on commodity hardware. A few major disruptions in tracking failures and accuracy degradation were observed in finger overlapping moments. Salz and Azam's (2019) virtual piano study had incorrect note assignments even from small positional errors. This highlighted the requirement of a precision threshold for such systems. These studies confirmed the limitations of single-viewpoint tracking systems, which are not captured by controlled dataset benchmarks.

For real time feedback pose estimation system, temporal smoothing techniques have been applied to reduce landmark jitters. A study done by Lugaesi et al. (2019) used the MediaPipe framework to reduce computing power by not using the traditional frame-by-frame approach. The study used landmark calculations via prior frames, meaning only the algorithm was lightweight. This was especially better for high-motion situations requiring coordinate stabilization. One drawback was in cases of very unexpected prior frames with rapid movement; unexpected delays were observed. Moments where the piano key was unpressed, redetection added measurable latency, causing inaccuracy in model prediction. Research on adaptive detection using prior frames could be explored rather than relying solely on per-frame identification to address rapid finger transitions.

The architectures handling hand tracking using a single camera face problems with occlusion. For research purposes, Hürst and Van Wezel (2013) designed a system that works with gesture-based models. They concluded that self-occlusion and finger overlaps caused the most misclassification. Multiple testers reported experiencing confusion, and accuracy drops during events with finger overlaps. Recent architectures utilizing the MobileNet backbone have tried discarding landmarks below a certain threshold to combat false positives. This approach introduced tracking gaps while eliminating false positives in occlusion events. Pu et al. (2023) assessed the HPE model's robustness under occlusion and rapid motion, revealing real world events reduce landmark detection more than what controlled experiments show. Methods for interpolation or prediction of these gaps could be further expanded upon for piano performances where fast music pieces with a lot of finger overlaps are main occlusion sources. Some improvements like implementing a predictive interpolation algorithm for handling occlusion, could certainly improve tracking rate for most musical pieces and applications.

3. Review of technologies

To conduct and produce meaningful research results, a combination of hardware and software has been picked precisely per requirement. The following sections below are some of the hardware devices used

3.1 Major Hardware Components

- Casio CTK-2400 electronic keyboard: this electrical piano provides a USB connection to any host machine using a printer cable. This allows MIDI signals to be read using python library Mido, which can be used to sync timings and MPJPE calculations. With the 61 keys, enough length for two-handed capture is possible.



Fig 3.1.a: CTK-2400 electrical piano



Fig 3.1.b: USB 2.0 to USB 3.0 cable

- Mobile camera connection through USB Type-C: The Google Pixel 7 was used via webcam option from the USB debug setting. The smartphone sensors can also be used to determine the illuminance of the environment.



Fig 3.1.c: Google Pixel 7 and USB Type-C cable

- Adjustable Camera Tripod: A camera tripod with a mobile phone adapter mount can ensure fixed recording height and angle, reducing any bias in data processing later.



Fig 3.1.d: Velbon adjustable camera tripod

- Host machine: The laptop used for developing the system and data processing is a Msi gaming laptop with Intel i-7 14th gen processor with NVidia RTX 5070 mobile chip with 16 GB RAM. This hardware enables fast data processing and model execution even in live video preview with the help of Nvidia CUDA support in Python libraries.



Fig 3.1.e: MSI gaming laptop

3.2 Software Components and Libraries

- Python 3.10+: The previous version of Python is used here due to the libraries not updated to work with the latest Python releases as of March 2026. Python 3.10 enables support for OpenCV and MediaPipe, making it the better choice for using a virtual environment. With support for machine learning libraries, good documentation, and CUDA performance, Python is chosen.
- NumPy: A Python library that can process thousands of data points in a structured format efficiently, in the case of this research, the midi timings and midi signals along with MPJPE calculations
- Pandas: A Python library where data can be put into a certain structure for easy retrieval and storage
- OpenCV: The cv2 Video capture allows retrieval of frames through USB connectivity and allows distortion fixes for individual frames. This is especially important as OpenCV also provides metrics for frames per second and latency to avoid a <20fps risk
- Mido and python-rtmidi 1.5.0: Mido is an opensource python library that provides low-latency MIDI signal capture using a high-level language.
- Matplotlib: Visualization of final data using graphs to show the impact of various variables considered while collecting data
- Git: Enable safe work version control in case a revert is required
- MediaPipe Hands: Actual testing framework for hand detection.
- OpenPose Hands: Second testing framework for hand detection.

4. Methodology

For this research, this section mentions and explains why this Rapid Application Development (RAD) was chosen over waterfall and agile due to its nature of iterative data collection pipeline development and flexibility. Also explains why RAD is better than others and the project's timeline.

4.1 Methodology description

Rapid Application Development (RAD) methodology was chosen for the development of the data collection system for this research. Martin (1991) defined the RAD as the development of rapid, time-boxed cycles of requirements analysis, prototype construction, user feedback, and revision. A fully completed specification is not required upfront, and RAD prioritizes development by understanding the requirements and system, and testing fundamental functional requirements first.

4.2 Research design plan

The Gantt chart starts with the initiation and risk planning of the research in December and ends with the research presentation in the month of May, totaling a time of twenty weeks. After allocating the time for hardware testing and software setup, the main data collection system development starts. The project timeline and the Gantt chart display Phase 2 (Weeks 5–8), the rapid development of hardware sync, data collection pipeline, and real-time overlay system creation over the span of four weeks. Since the hardest part of the research is to collect enough data to produce meaningful results, the main data collection phase, Phase 3 (Weeks 9–12), has been prioritized to be done in the span of about three weeks. This phase is to collect data following the ethical considerations posed by the university and signed contracts, which introduce user identity safety. In Phase 4 (Weeks 13–16), the analysis phase compares MPJPE, secondary metrics, and overall statistics along with result visualization using graphs. Phase 5: producing a meaningful report that spans from the beginning to the end of the research, tabulates result variables, and outputs research-backed feedback for vision-based models and produces a new dataset governing vision and music instrument interaction contribution for the open-source community, and ends with a conclusion that can be further explored via other research and projects.

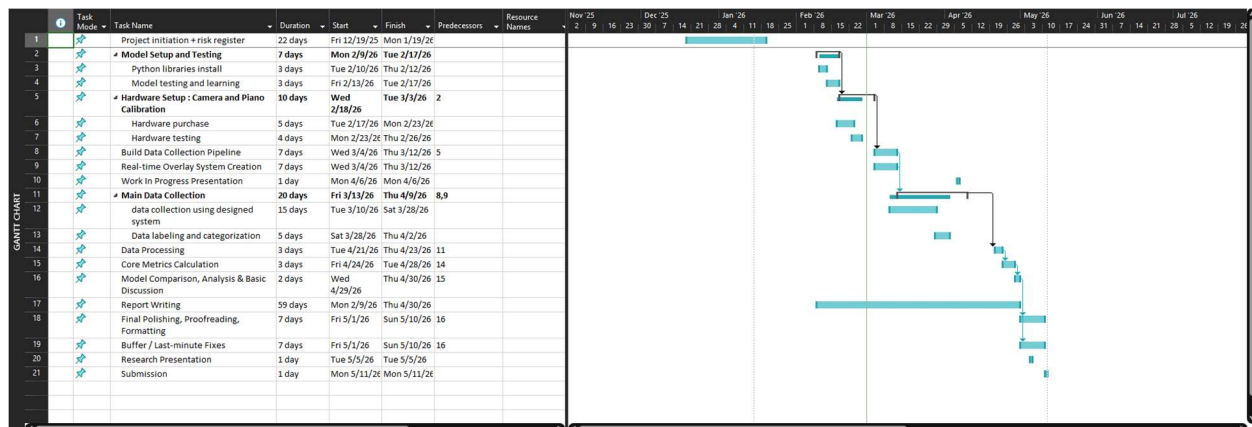


Fig 4.2.a: project timeline and Gantt chart

4.3 Methodology defense

Several methodologies were proposed at the beginning of the project initiation. Before RAD was considered, the waterfall methodology was reviewed. The waterfall methodology expects a linear approach and allows no time for iterative refinement, recursion, and reverting in case of system failure. The research expects several experiment protocols to be fulfilled, including playing specific patterns, computation of MPJPE and failure detection, vision models, landmark estimation, and adjustment pilot testing. Waterfall methodology would have introduced unnecessary overhead in case of these factors needing adjustment. The second methodology considered was Agile, which is highly suitable for team-based software development projects. It introduces multiple terms and events that suit more of a product development cycle than a single-person research project. Agile was therefore rejected due to its emphasis on delivering user value in a sprint situation, which does not translate to statistical analysis and dataset tuning. RAD offers an optimal balance as it enforces completing functional development quota to display progress, while improving practical discoveries during the research lifetime. The flexibility provided by RAD methodology allowed valuable insight into MIDI timeline synchronization with video data timeline issues to be diagnosed and discussed.

5. Research/ Product design

The product design for this research is the data collection pipeline. The prototype requires a mobile phone positioned on top of the piano, where the MIDI data is collected through a USB connection and is synchronized with the video data. Offline processing is done using MediaPipe and OpenPose models and produces per-session result files that directly display the outcomes presented in Section 7 results. The functional and nonfunctional requirements decided in the initial project plan are also summarized and show what was implemented, how it was implemented, and what wasn't.

5.1 Data collection pipeline system

Setting up the data collection pipeline from an upward angle with almost a ninety-degree view is the initial setup for open-source models to track finger movement.



Fig 5.1.a: Data collection pipeline setup

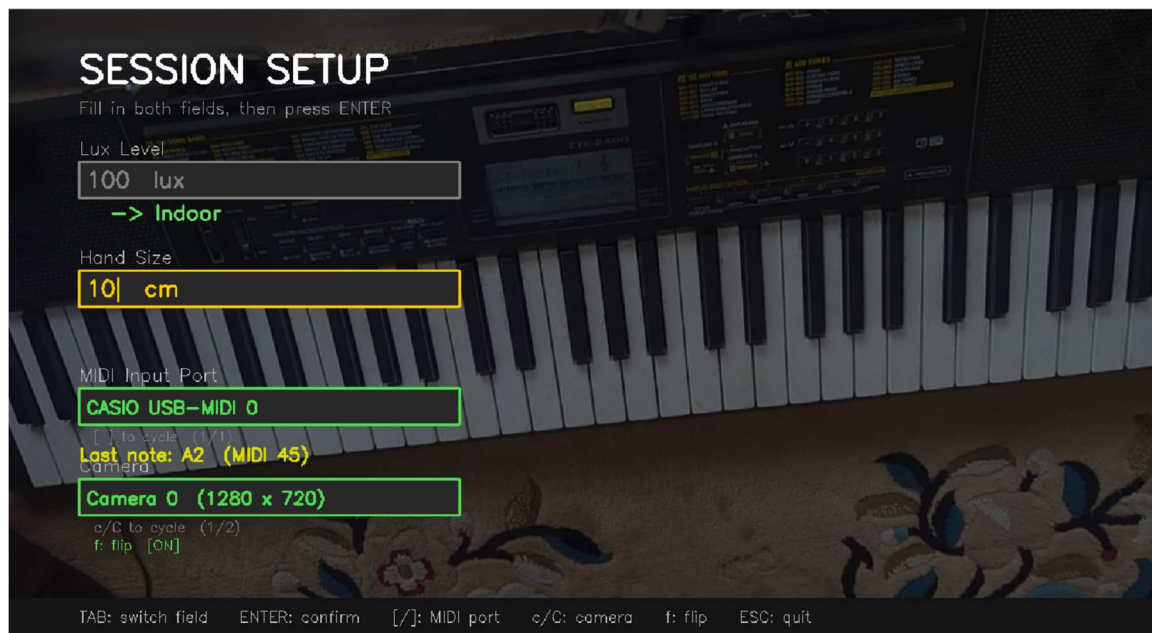


Fig 5.1.b: Session setup parameters

System Architectural Diagram - Piano Finger Tracking



Fig 5.1.c: System Architecture Diagram

Use Case Diagram - Piano Finger Tracking System



Fig 5.1.d: Use Case Diagram

6. Implementation and Testing

Before data collection was started, the features were developed, such as finger landmark tracking using polygon-based containment and center-based MPJPE calculation, making for a double cross-check for model performance. Also, MIDI synchronization with video was the most important feature to work without any errors to produce accurate research data before claiming any outcomes. A live preview was developed first and tested to ensure the offline processing works in the same way. Also, this section mentions how the participants' categorization and lighting conditions were checked.

6.1 Dataset Collection Pipeline Overview

The custom dataset collection pipeline has multiple features. The pipeline initialized a session where participants' data categorization is done automatically, an interface with a mask aligned to the piano keys marking the border and the center of the key to establish ground truth, midi streams, and video data are captured simultaneously for later offline MPJPE analysis. All participants' data is saved under data/raw/p<id>, meaning no other personal details are stored, ensuring data confidentiality and anonymity.

6.2 Participant and Environmental Characterization

For cross-participant analysis, two variables were systematically logged before recording started. The first variable was skin tone categorization according to Fitzpatrick phototype. The second variable was ambient illumination measured by phones lux meter. The potential of both variables affecting landmark detection confidence in camera-based models was identified as a potential confound.

6.2.1 Skin Tone Classification (Fitzpatrick Scale)

The skin tone classification was done automatically using the colorimetric method (Chardon et al., 1991) using the individual typology angle (ITA). Without any external identification models, this method provides an independent skin reflectance measurement. CIELab pixel values from the palm region, which are MediaPipe landmarks 0,1,5,9,13 and 17, are sampled. The resulting ITA score is correlated with the corresponding Fitzpatrick scale category (Type I-VI). The resulting value is computed with 20 frames average.

The ITA score was computed as:

$$\text{ITA} = \tan^{-1} \left(\frac{L^* - 50}{b^*} \right) \times \frac{180}{\pi}$$

Where:

L^* : perceptual lightness

b^* : the yellow–blue opponent channel

The Fitzpatrick Scale

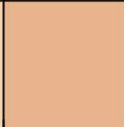
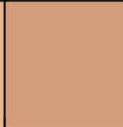
						
Type	I	II	III	IV	V	VI
	Light, Pale White	White, Fair	Medium White to Olive	Olive Tone	Light Brown	Dark Brown

Fig 6.2.1.a: FitzPatrick scale (FitzPatrick, 1988)

6.2.2 Ambient Illuminance Classification

The Illuminance is classified by **Dim** (<100 lux), **Indoor** (100–499 lux), and **Bright** (≥ 500 lux). Rather than automatic measurement, manual entry was chosen over automatic measurement through a smartphone app displaying sensor data, to avoid dependence on additional hardware. This enables sufficient granularity for landmark detection performance under lighting variations. Lee et al. (2023) demonstrated significant accuracy loss under low illuminance for pose estimation models. In landmark detection, light levels should be considered a primary variable.



Fig 6.2.2.a: LUX Meter Mobile Phone (14.67 lux)

6.2.3 Hand Size Classification

The hand sizes were only kept as metric, meaning different hand sizes weren't placed under any groups. The hand sizes are in centimeters and measured from palm to palm with a measuring tape.

6.3 Pipeline MPJPE Calibration and Bias Reduction

When developing the dataset collection pipeline, the 2D calculation of MPJPE introduced systematic bias due to the natural anatomy of human fingers. The fingers with the shorter distance were more towards the designed center mask of the piano overlay, causing bias to longer fingers playing the same or different key. The second metric that introduced bias was the piano keys being longer on the y-axis. When playing a selected note, if another finger besides the actual finger playing the note was nearer, a wrong flag was marked for artificially inflated MPJPE.

Results: <25% of the white key width (approximately 10px at the calibrated resolution) were classified as correct detection of landmarks

6.3.1 Horizontal MPJPE: Fingertip-to-Key Centre Offset

To address the first metric causing bias, the 2D MPJPE calculation was changed to horizontal-only displacement, removing the bias produced by natural finger depth. This makes the entire vertical distance of each key the center rather than just one point on the key.

Single-note horizontal error:

$$E(x) = |x_{tip} - x_{key}|$$

Where:

x_{tip} : predicted fingertip horizontal coordinate

x_{key} : horizontal center of the ground truth key

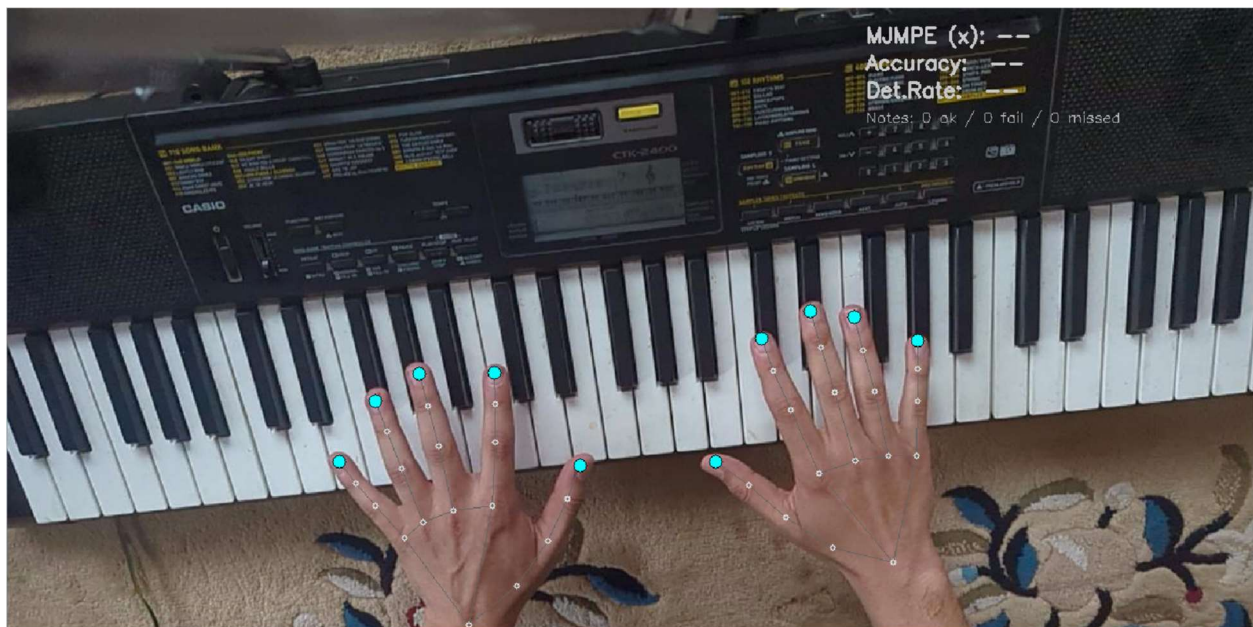


Fig 6.3.1.a: MediaPipe hands landmark preview

6.3.2 Polygon Containment for Finger Detection

The second metric causing bias is the 2D centroid proximity finger to perspective polygon containment, meaning the fingertip lying within the physical bounding polygon is the note-playing finger. The solution is to implement a visual mask marking a key's boundary, which can be used to identify false flags as well.

When testing the initial system for dataset collection, a systematic misclassification was identified. Due to the configuration of the polygon mask implementation, half for the left hand and half for the right hand caused incorrect assignments of finger notes. This was

improved upon by using MediaPipe's native handedness output, which gives left and right as output using landmark geometry (Zhang et al., 2020). This made the system model-driven rather than screen-position dependent.

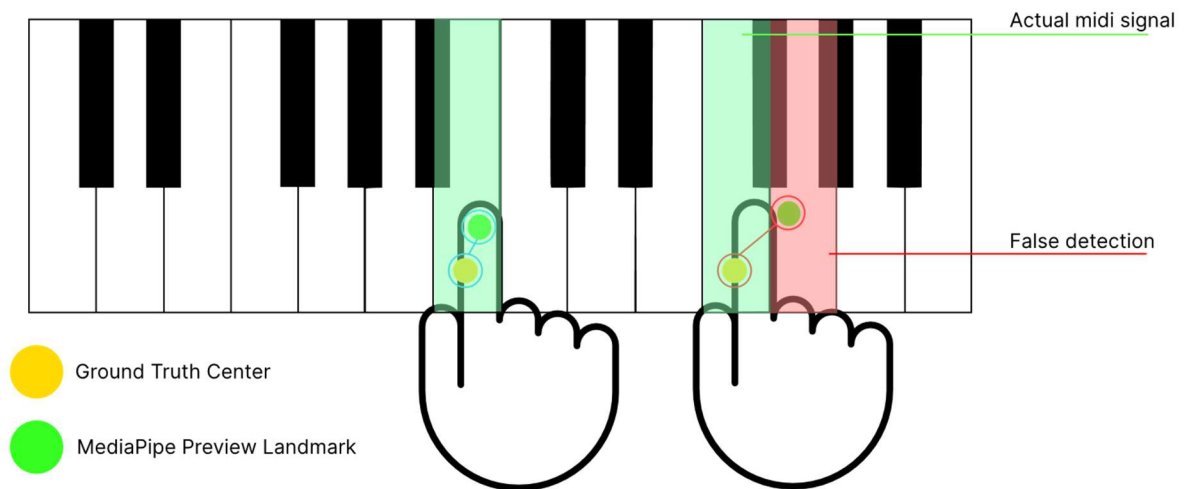


Fig 6.3.2.b: polygon containment mask

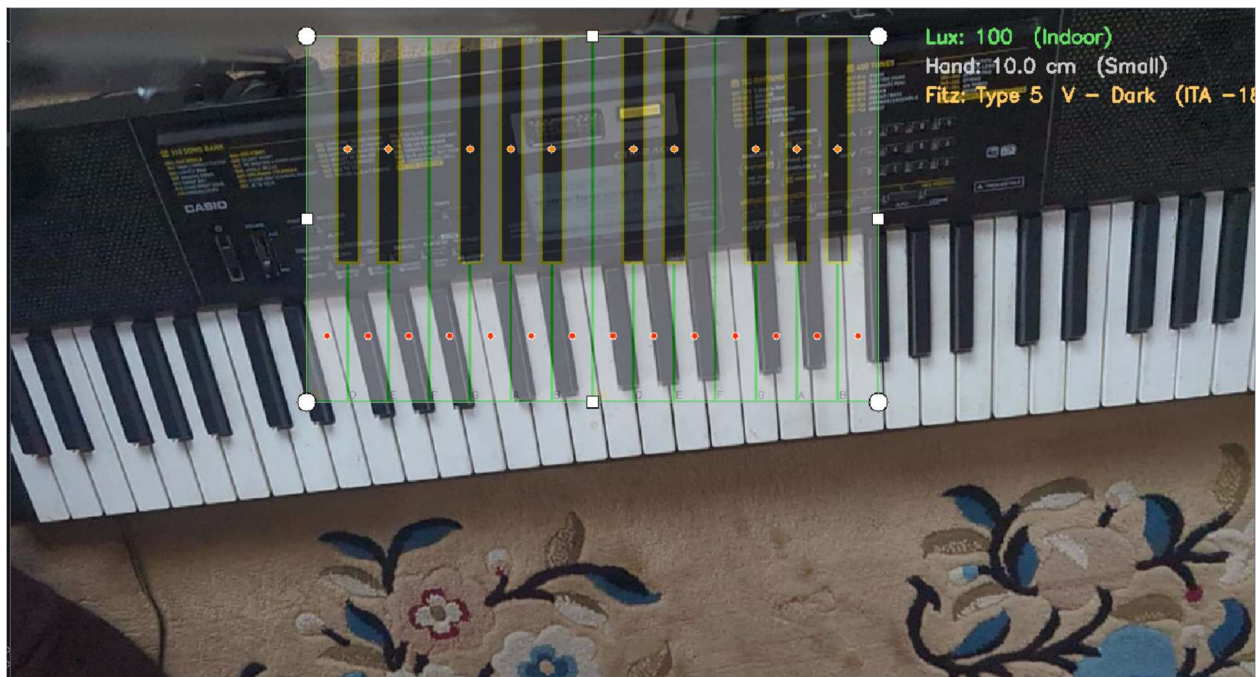


Fig 6.3.2.c: polygon containment mask preview

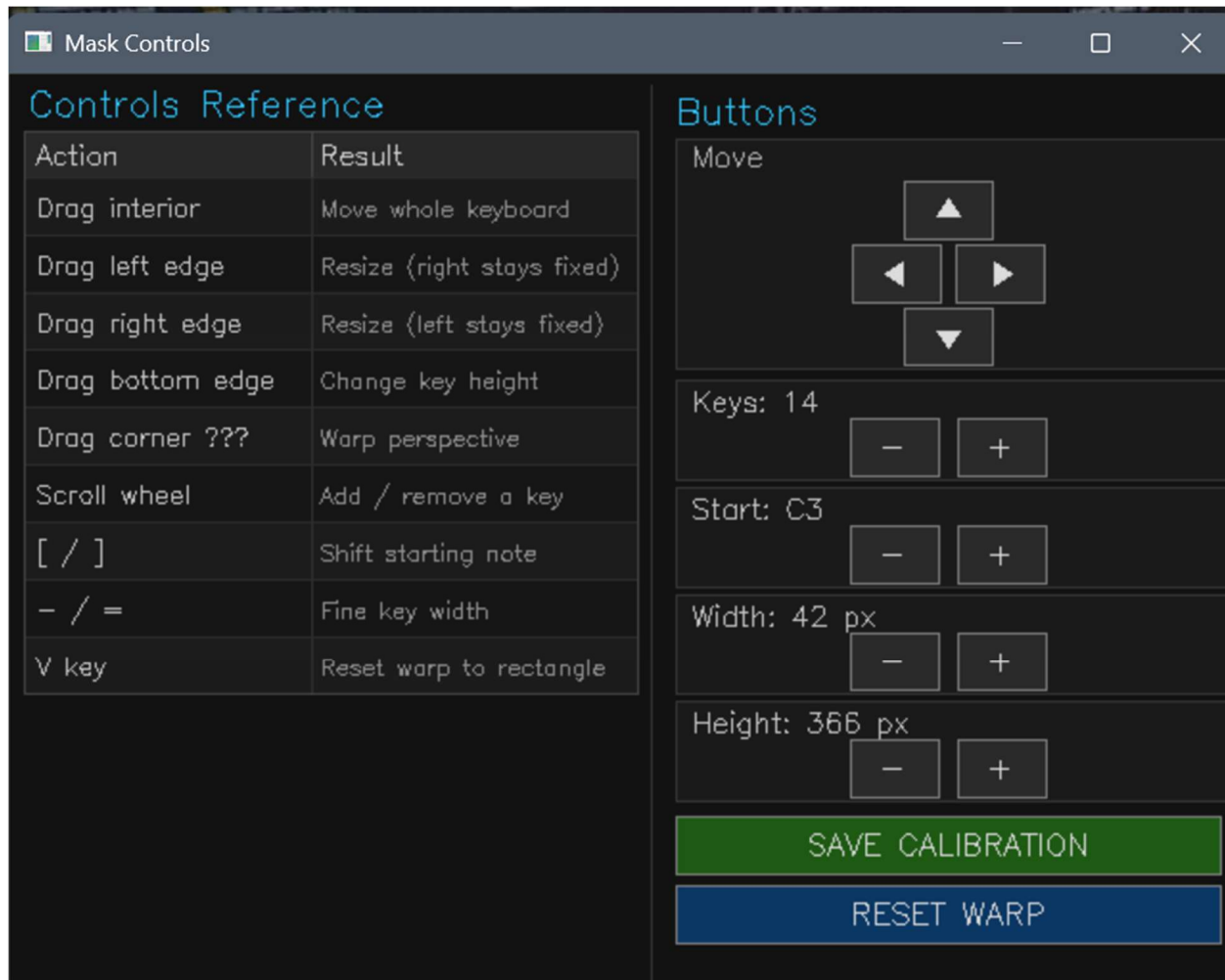


Fig 6.3.2.d: Polygon containment mask controls

6.3.3 Perspective Homography Calibration

Additionally, due to the tripod placement rarely producing a perfect straight view of the mask to perfectly sync with the live preview and the offline calculation, a perspective homography transformation matrix calibration, which warps the mask to a quadrilateral fitting the corners with corner drag controls, was developed following the approach outlined in Siddique et al. (2019). The `cv2.getPerspectiveTransform` computed the homography matrix. The transformation is defined as:

$$\tilde{p}' = H\tilde{p}, \quad H \in R^{3 \times 3}$$

Where: $\tilde{p} = [x, y, 1]^T$ are homogeneous coordinates

The warped vertex computation:

$$v'_i = \pi(Hv \sim i)$$

The warped polygon vertices are recovered via dehomogenisation:

$$\pi \left(\begin{bmatrix} x \\ y \\ w \end{bmatrix} \right) = \left(\frac{x}{w}, \frac{y}{w} \right)$$

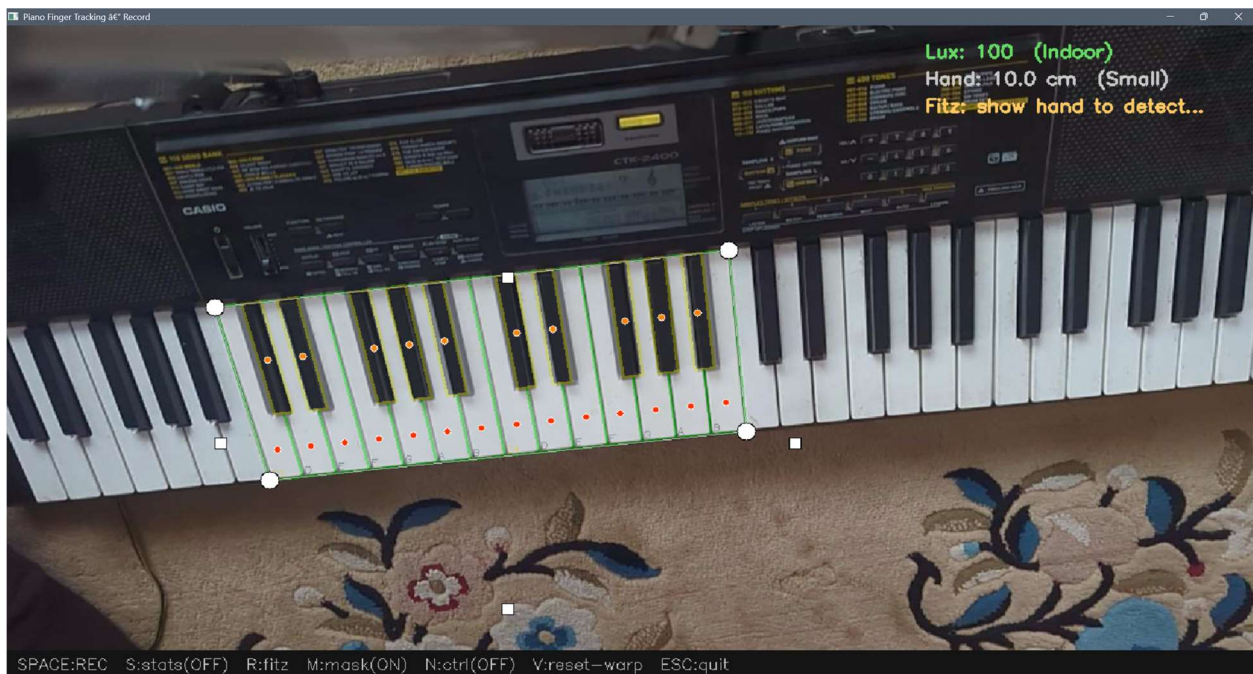


Fig 6.3.3.a: homography calibration mask

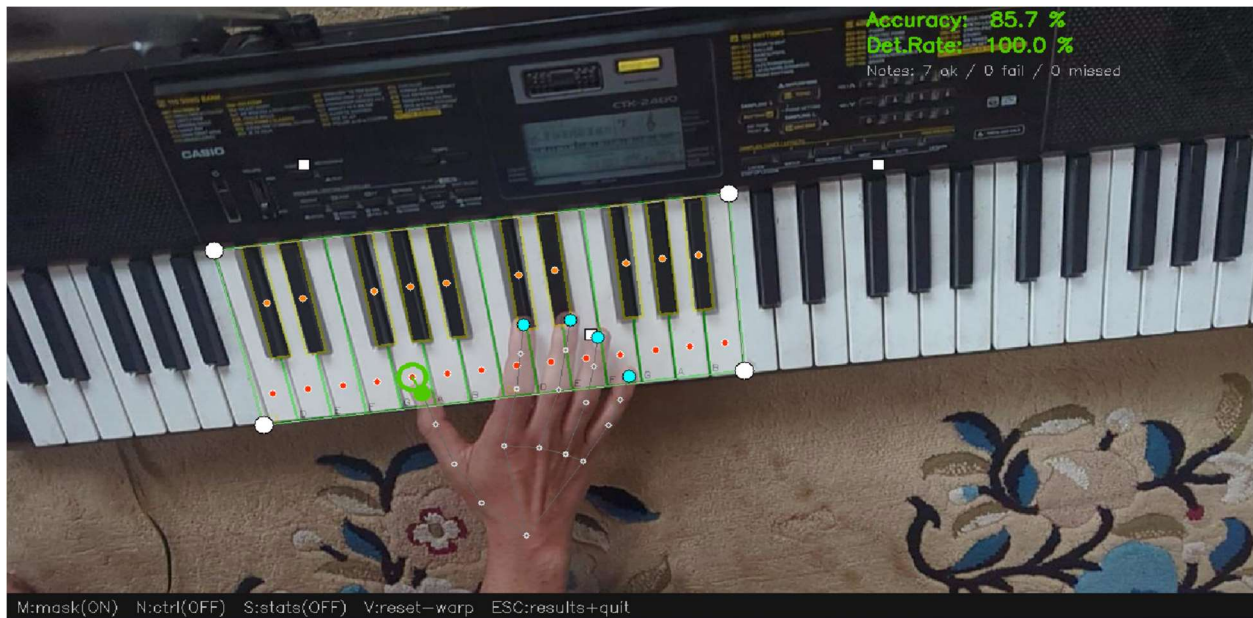


Fig 6.3.3.b: correct landmark detection

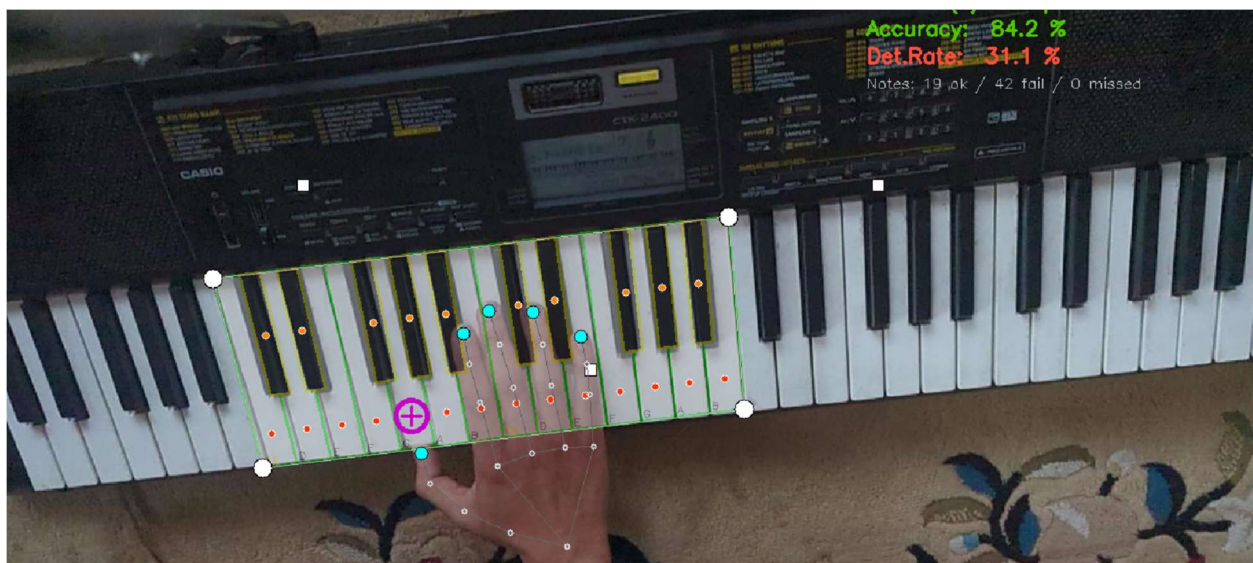


Fig 6.3.3.c: out of polygon mask false detection



Fig 6.3.3.d: mask visualization after accounting for vertical finger depth

6.4 Synchronized Multi-modal Data Acquisition

For accurate MPJPE calculations, MIDI note events and video frames require sub-frame temporal precision. By utilizing a shared time origin anchor by `time.pref_counter()` call when the recording starts, the frame and MIDI timestamps are also recorded independently.

6.4.1 MIDI–Video Temporal Alignment

MIDI recorder runs as a background daemon thread. Every 1ms, using `mido's iter_pending()` interface MIDI input port is surveyed. Timestamped messages have a shared origin (`rec_t0`). The frame logger simultaneously records (`frame_index`, `time_s`) for every frame written to disk. Two aligned time series are saved in CSV and JSON files with the same origin expressed in seconds. By using a linear search on the log, the nearest note event to the video frame is retrieved. This ensures each note corresponds to the closest captured frame.

6.5 System Performance and Detection Statistics

Frames per second (FPS), MediaPipe inference latency (ms), and frame processing latency (ms) are some metrics calculated in real time to monitor performance. To identify any performance degradation, these metrics are displayed live during recording.

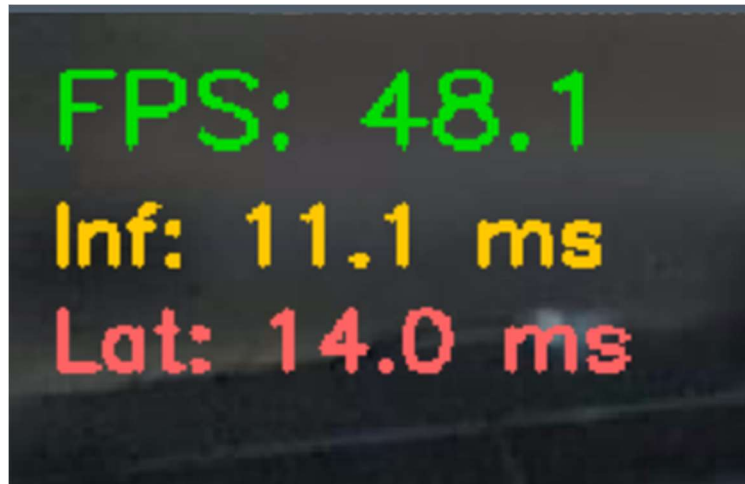


Fig 6.5.a: live_preview statistics

For each session, detected statistics for every MIDI note_on event are categorized into one of three outcomes: matched (correct landmark detection inside key polygon, detection fail (no fingertip landed within the key boundary when hand landmarks were present), or missed (no hand landmarks were detected). From the outcomes, these metrics were obtained:

The Mean Per Joint Position Error (MPJPE):

$$\text{MJMPE} = \frac{1}{N_{\text{matched}}} \sum_{i=1}^{N_{\text{matched}}} |x_{\text{tip},i} - x_{\text{key},i}|$$

the Detection Rate (DR):

$$\text{DR} = \frac{N_{\text{matched}}}{N_{\text{matched}} + N_{\text{fail}} + N_{\text{missed}}} \times 100$$

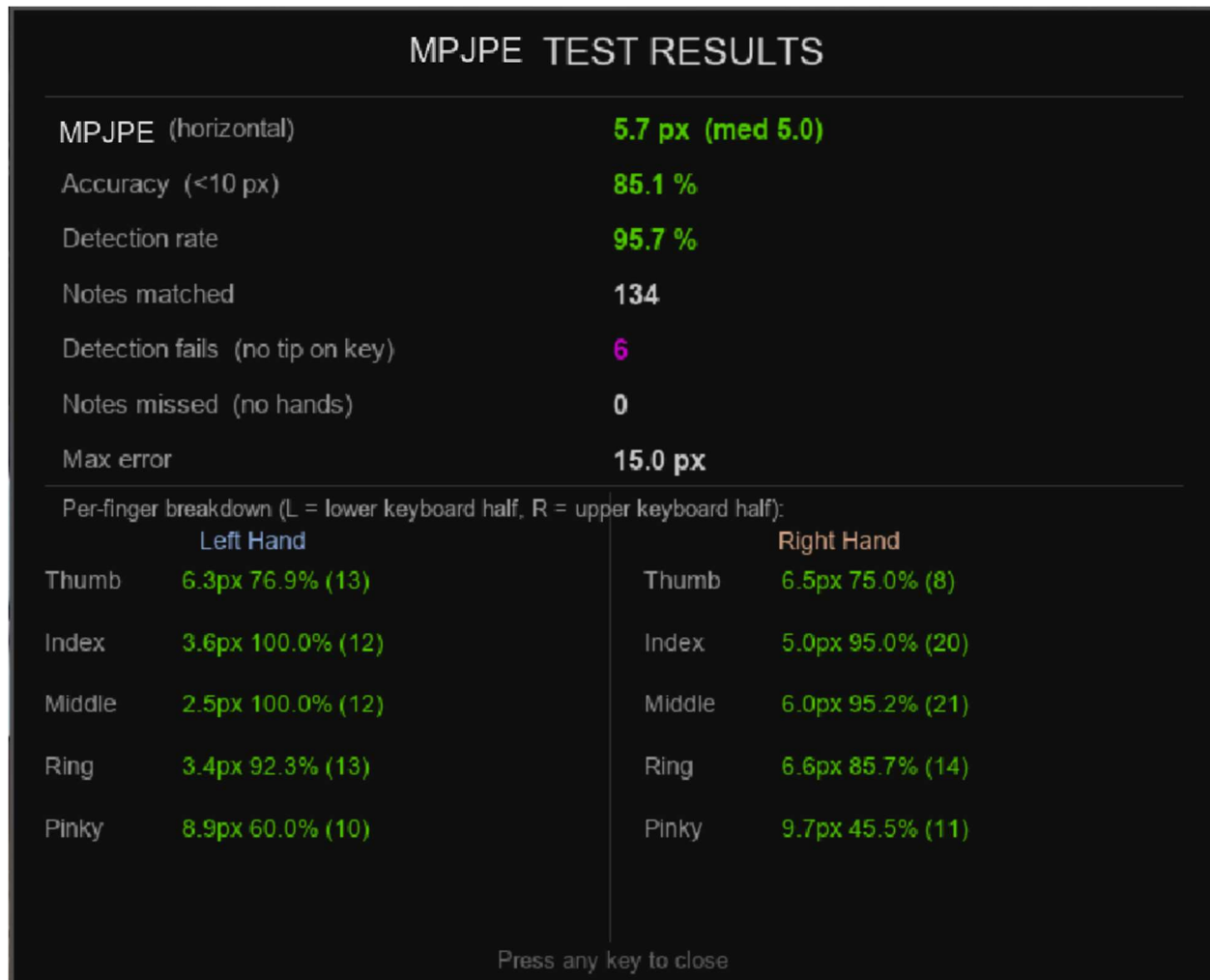


Fig 6.5.b: live_preview.py per hand per finger result

6.6 Participant Study Design

A study with three pre-planned patterns was completed by each participant using the same hardware configuration to evaluate tracking performance among participant demographics and illuminance conditions.

6.6.1 Participant Recruitment and Inclusion Criteria

Participants' recruitment was done through recommendations and signing consent forms. No exclusion criteria were applied based on skin tone or hand size. For relevant tracking performance diversity across Fitzpatrick phototypes and hand sizes was actively sought. A total of 25 participants participated in this study, where some participants tried on various lighting variations, equating to 38 videos.

6.6.2 Session Protocol and Procedure

Each participant completed a predefined pattern session. After manually entering the lux value, hand size, and identifying participants, the following three patterns were played.

1. Five-Finger Position: C-D-E-F-G pattern for both hands with no finger crossing, ascending and descending 4 times.
2. One-Octave C Major Scale: C to C on the octave with one finger cross.
3. Black Key Pentatonic Scale: C#-D#-E#-F#-G# on the black keys ascending and descending 4 times

6.6.3 Dataset Structure and Storage

All raw data from each session was written to a dedicated folder at data/raw/p####/. Each participant folder has frames.csv containing all the frame data with the timestamps, the details of the session with participants' variables and illuminance data, and midi data in JSON and a mid-file with frame and timestamps synchronized.

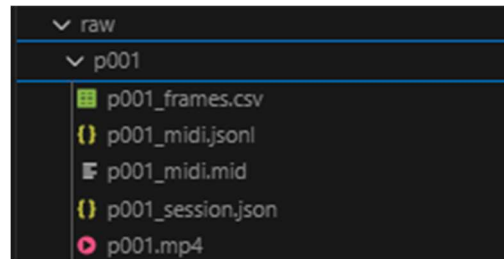


Fig 6.3.3.a: Raw data saved structure

6.7 Offline MPJPE Analysis

After the data collection process, the MIDI data and frame data were processed by both MediaPipe Hands and OpenPose independently. The resulting MPJPE, detection rate, and per-hand with per finger breakdowns are stored in the processed folder individually, meaning each participant will have their own result file, p#_mediapipe_results.json and p#_openpose_results.json.

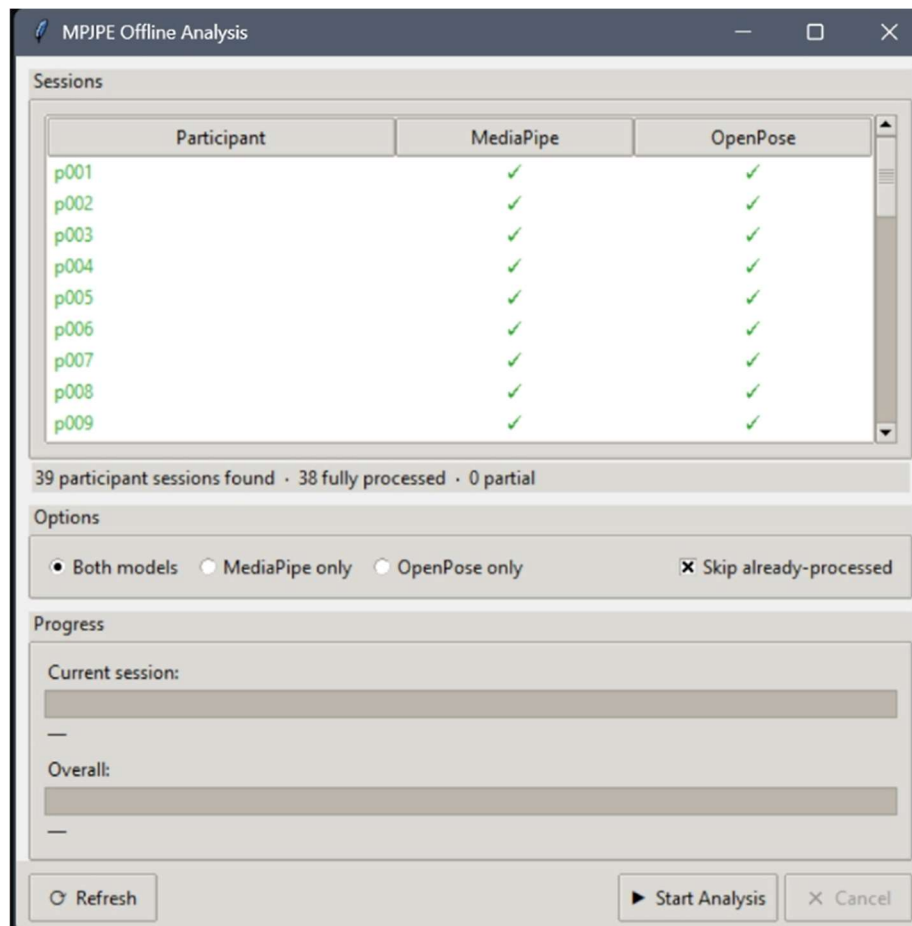


Fig 6.7.a: Offline processing UI (analyse.py -ui)

7. Product Evaluation

7.1 Functional Requirements Completion

The functional requirements established in the Initial Project Plan are summarised below in MoSCoW priority order with completion status.

#	Functional Requirement	MoSCoW	Status
1	Finger landmark tracking using vision models	Must	Completed
2	MIDI ground truth capture (Casio CTK-2400)	Must	Completed
3	Temporal synchronisation	Must	Completed

	between MIDI and video		
4	Support multiple open-source models	Must	Partially completed (MediaPipe and OpenPose; MoveNet excluded after pilot testing)
5	Camera-to-keyboard spatial calibration	Must	Completed (homography polygon mask)
6	Data logging system	Must	Completed (frames.csv, MIDI JSON, video)
7	Keypress event analysis	Must	Completed (polygon containment + nearest-fingertip)
8	Quantitative performance metrics	Must	Completed (MPJPE, detection rate, FPS, latency live-monitored)
9	Controlled variation of experimental conditions	Must	Completed (Fitzpatrick II–V, Dim/Indoor lux, 7.5–12.5 cm hand sizes)
10	Statistical comparison of models	Must	Completed (descriptive statistics, group means with $\pm$ SD)
11	Model selection justification	Must	Completed (Section 8)
12	Reproducible experimental protocol	Should	Completed (scripts and documentation released)
13	Evaluation under additional models	Could	Not completed
14	Real-time visualisation overlay	Should	Completed (live_preview.py)
15	Automatic occlusion event annotation	Could	Not completed
16	3D hand pose estimation	Won't	Correctly excluded

17	Mobile / embedded deployment	Won't	Correctly excluded
18	Multi-camera / multi-view tracking	Won't	Correctly excluded
19	Real-time audio feedback (gamification)	Won't	Correctly excluded

Table 7.1.a: Function Requirements Summary

7.2 Non-Functional Requirements Completion

The non-functional requirements established in the Initial Project Plan are summarized below in MoSCoW priority order with completion status.

#	Non-Functional Requirement	MoSCoW	Status
1	Research-focused functional prototype	Must	Completed
2	Fair and controlled comparison conditions	Must	Completed (identical camera, hardware, room conditions across sessions)
3	Python + open-source libraries	Must	Completed (Python 3.10+, OpenCV, MediaPipe, Mido, NumPy, Pandas, Matplotlib)
4	Ethical compliance and participant protection	Must	Completed (university ethics approval Stages 1 and 2, consent forms, anonymised storage)
5	Real-time performance $\geq 15-20$ FPS	Should	Completed (sustained 30 FPS in live_preview.py during recording)
6	Code quality, documentation, reproducibility	Should	Completed (organised script structure, SCRIPTS.md documentation, version-controlled via Git)
7	Basic robustness and error handling	Could	Completed (daemon-thread

			MIDI capture continues across hardware disconnects, graceful logging of errors)
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Table 7.2.a: Nonfunctional summary

7.3 Potential Benchmark Application

For further refinements, a benchmark application with buttons can be developed for ease of script changing instead of using the terminal for every functionality.

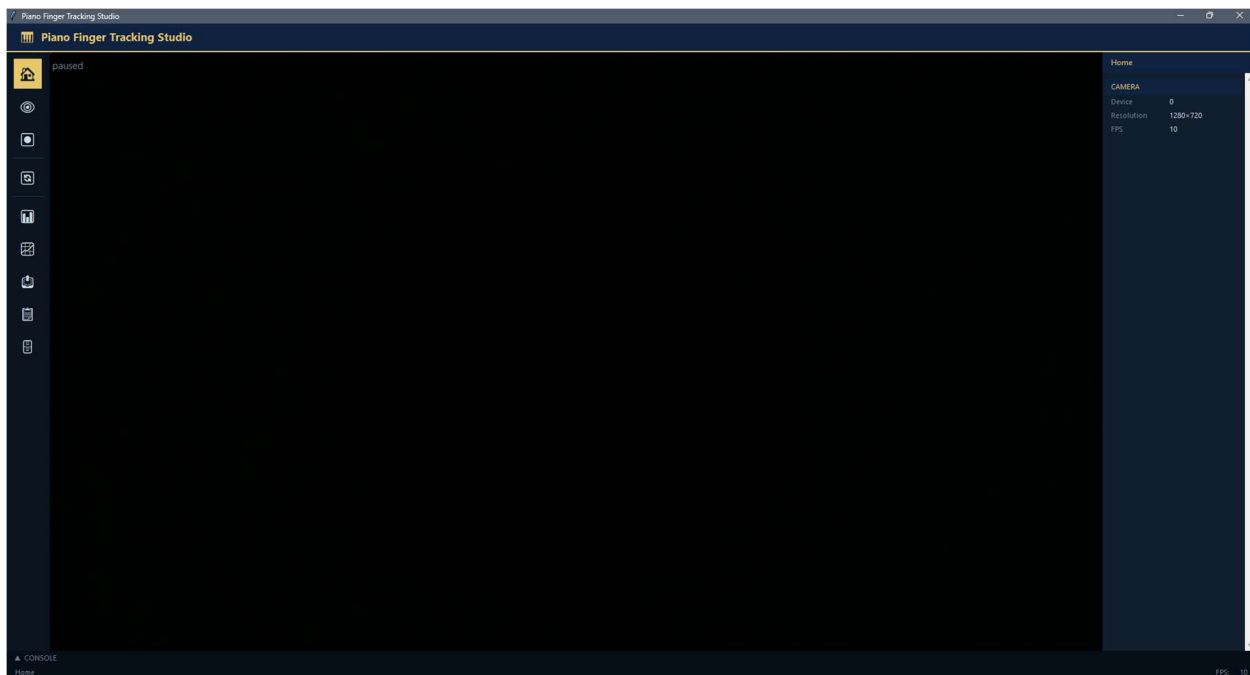


Fig 7.3.a: Prototype UI design benchmark app

8. Research Evaluation

Over 25 participants played the piano under the setup, totaling up to 40 videos. Some of the participants played twice under different lighting conditions dim (<100 lux) and indoors (>100 lux). The resulting data had x number of rows from 3.30+ GB of video data.

The below sections are derived from all the synchronized video and midi data processed thought the offline processing scripts analyse.py.

Participants comprised of Type II to Type V only for this research, where most of them were categorized into Type III and IV. For testing in conditions where most piano users play only in indoor lighting, dim lighting was tested, and most videos were recorded at

dim lighting. The mean hand size was 10.3cm; the data collected leaned more towards having sizes > 10 cm.

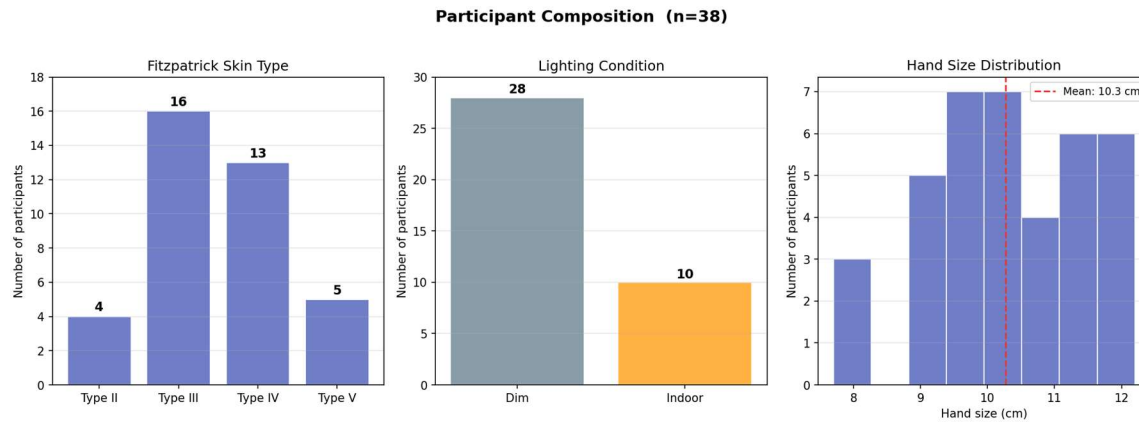


Fig 8.a: test data diversity charts

8.1 Overall Model Performance

8.1.1 Detection Rate

As mentioned from the previous formula in section 6.3.1, MPJPE is calculated according to the defined center points from the calibration mask. Detection is possible if the fingertip landmark is within the polygon boundary of the MIDI event. Since the OpenPose framework requires highly preprocessed, cropped images of hands, the model failed significantly in detecting hands from the video data. Simon et al. (2017) concluded that OpenPose is unsuitable for direct deployment without a video preprocessing stage, where the image is tightly cropped to where the hands are located.

MediaPipe matched the most events, 3017 events out of 4764, which equates to 63.7%, while OpenPose matched very few events due to its cropped hands requirement, 388 events, which equates to 8.1%. The bar graph visualizes the outcome as proportional stacks per model, and the table summarizes the data shown with exact numbers

Summary table for model performance:

Outcome	MediaPipe (n)	MediaPipe (%)	OpenPose (n)	OpenPose (%)
Matched	3017	63.3	388	8.1
Detection fail	1365	28.6	1072	22.5
Missed (no hands)	382	8.0	3304	69.3
Total	4764	100.0	4764	100.0

Table 8.1.1a: Model Summaries

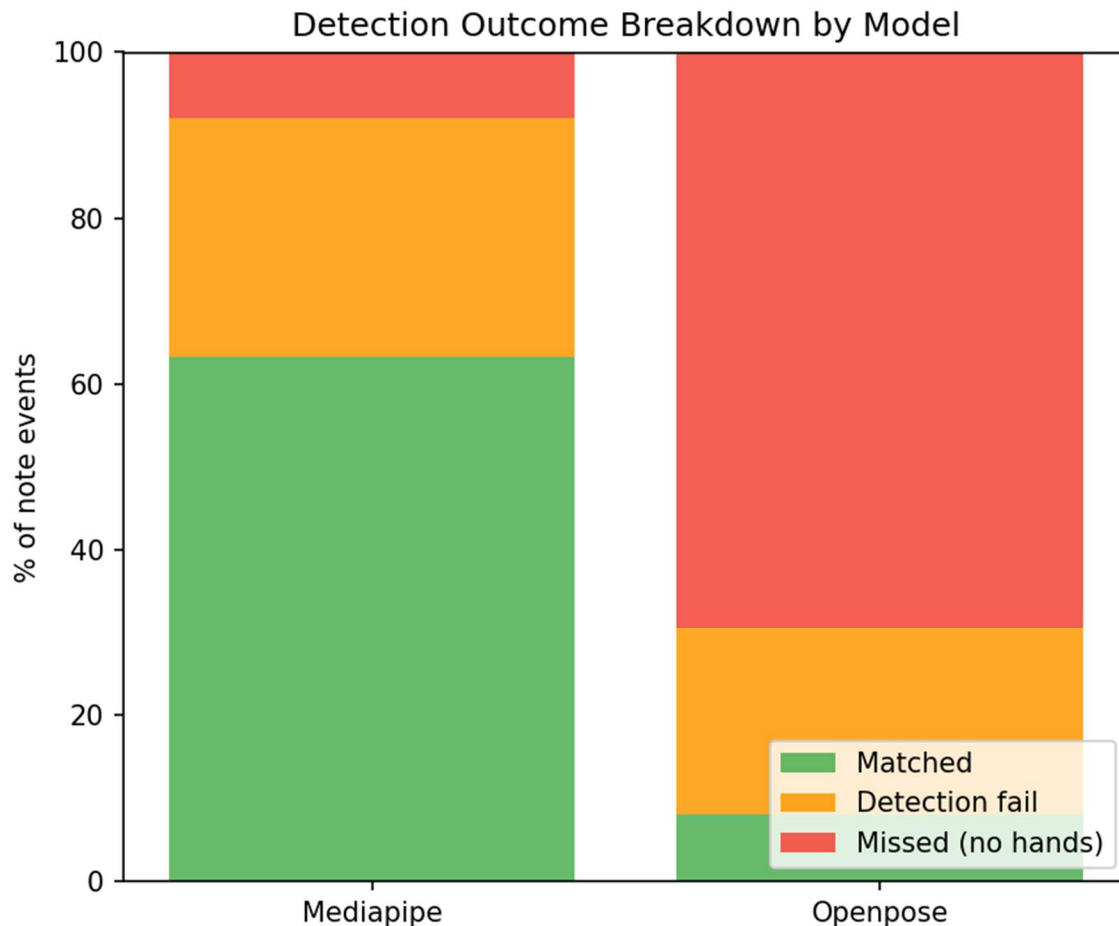


Fig 8.1.1.a: MediaPipe vs OpenPose detection rate chart

8.1.2 Hand Assignment and Per-Finger Computation

At first, to separate hands and fingers on the generated dataset, MediaPipes L and R labels were used, and each landmark's labels were also converted to data. MediaPipe's multi_handedness classifier was used for the first model. OpenPose didn't provide a similar feature, so both hands' results were classified as one hand. Now, each hand was classified as left and right for Mediapipe, with the geometric thumb-MCP / wrist x-coordinate check correcting any palm-down mirror flips. At first, hand assignments through keyboard position relative to frame midpoint were created to assign L and R for OpenPose, but the large number of detection failures might skew the results and misclassify L and R as well. Since this feature only decides which side the physical hands lie on the frame and the keyboard, both hand results were combined and displayed as one for OpenPose. Therefore, OpenPose's per-hand split is treated as unreliable for this research and must not be used to support the L and R claims. Both models used indices 4,8,12,16 and 20 for Thumb, Index, Middle, Ring, and Pinky, respectively, to identify

fingertips. Per-hand match rate and per-finger MPJPE were calculated using the formulas mentioned in sections 6.5 and 6.3.1 for a single hand or single finger.

8.1.3 Per-Hand and finger detection and match rate

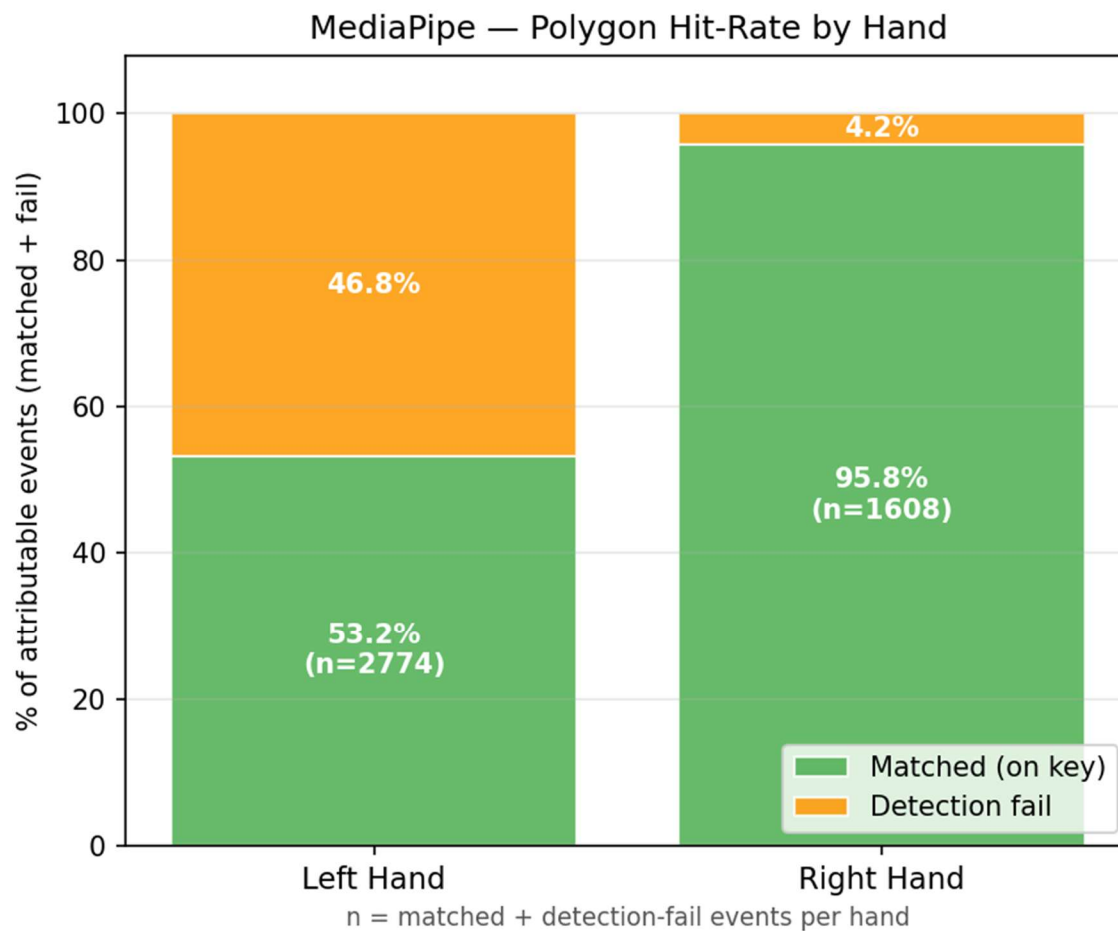


Fig 8.1.3.a: Per Hand and finger detection rate for MediaPipe

Metric	Thumb	Index	Middle	Ring	Pinky
Within threshold (%)	85.7	85.1	83.8	82.0	87.4
Above threshold (%)	14.3	14.9	16.2	18.0	12.6
Total matched events (n)	342	807	757	689	422

Table 8.1.3.b: Per finger match rate MediaPipe

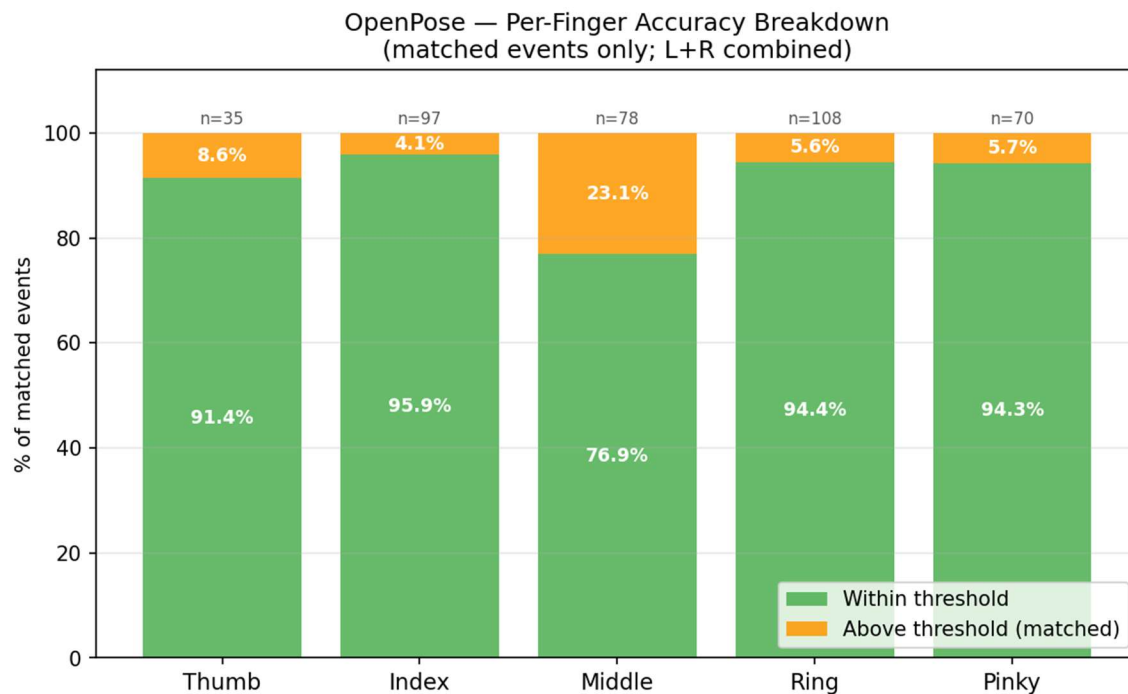


Fig 8.1.3.c: Average Finger detection rate for OpenPose (Low n detection rate)

Metric	Thumb	Index	Middle	Ring	Pinky
Within threshold (%)	91.4	95.9	76.9	94.4	94.3
Above threshold (%)	8.6	4.1	23.1	5.6	5.7
Total matched events (n)	35	97	78	108	70

Table 8.1.3.d: Average Finger detection rate for OpenPose Summary

8.1.4 Detection-Fail Rate by Fitzpatrick Type

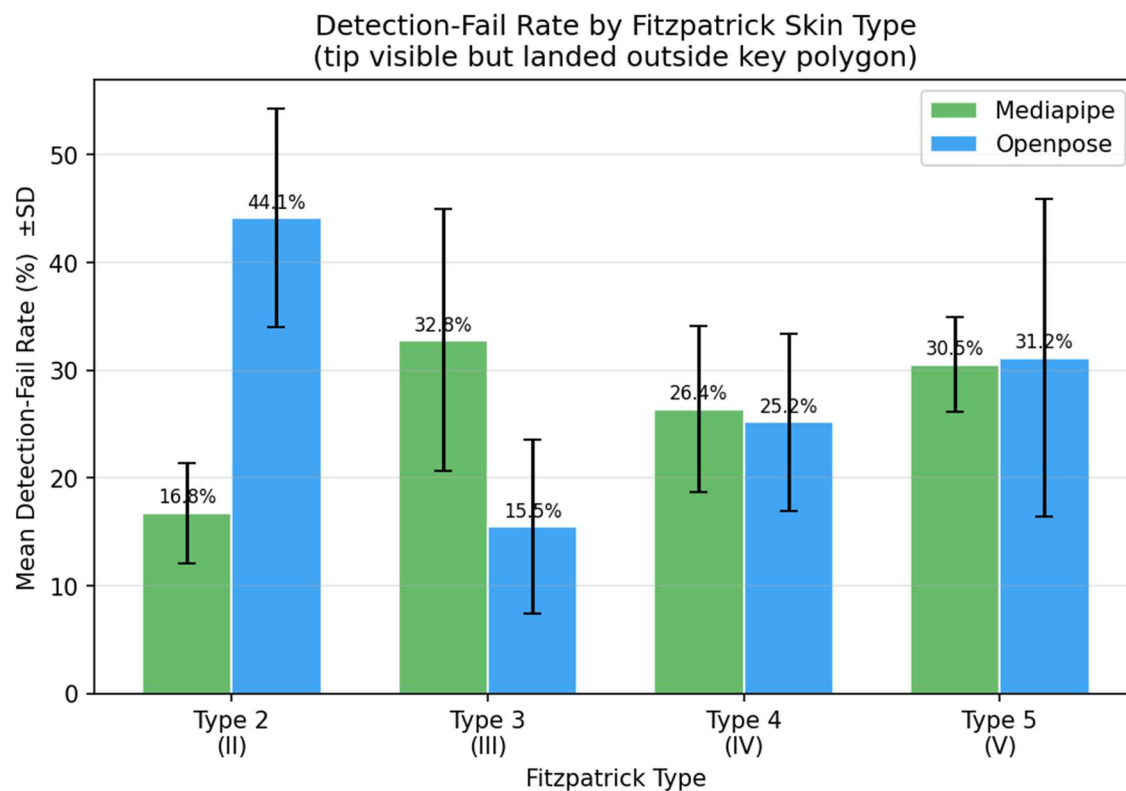


Fig 8.1.4.a: detection fail rate by skin tone

FitzPatrick scale	MediaPipe (DFR %)	OpenPose (DFR %)
Type II	16.8	44.1
Type III	32.8	15.5
Type IV	26.4	25.2
Type V	30.5	31.2

Table 8.1.4.b: detection fail rate by skin tone

8.1.5 Detection-Fail Rate by Lighting Condition

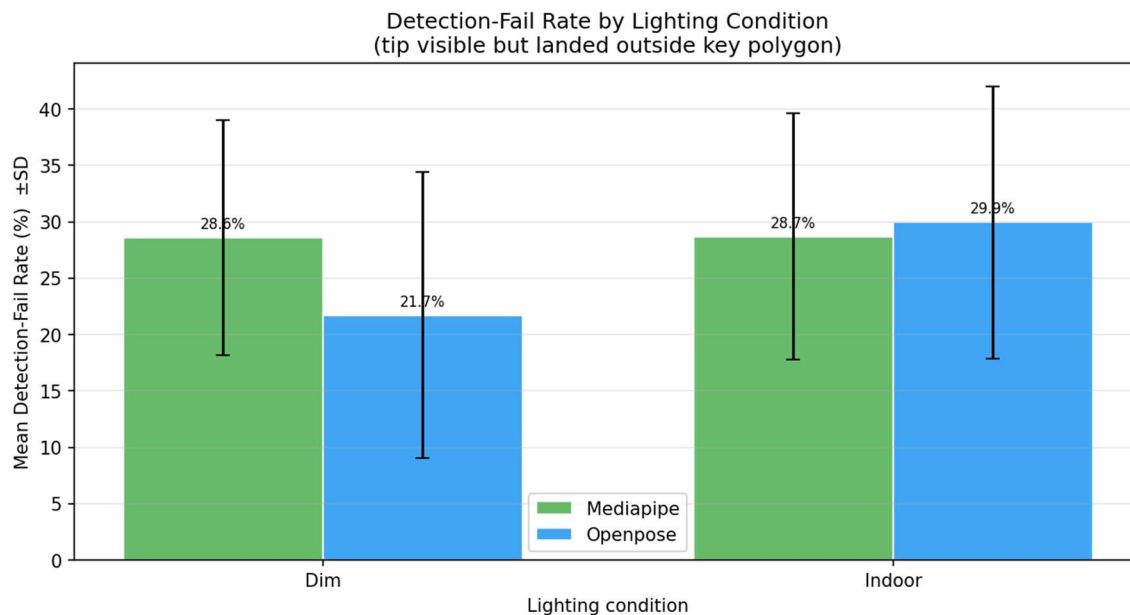


Fig 8.1.5.a: detection fail rate by lighting condition

Lighting condition	MediaPipe (DFR %)	OpenPose (DFR %)
Dim	28.6	21.7
Indoor	28.7	29.9
Bright	—	—

Table 8.1.5.b: detection fail rate by lighting condition summary

8.1.6 Detection Outcome Breakdown

For quick performances with multiple fingers overlap and live demand for landmark processing, MediaPipe bar far excels in performance in comparison to the OpenPose framework. Even if OpenPose slightly outperforms MediaPipe in terms of the MPJPE metric, this is a case of selection bias. The better MPJPE results of OpenPose are only based on about 8% of the detections, while 92% of them were never detected to produce a landmark result.

While both models failed to exceed 80% detection, MediaPipe detection failure was only 28.7% of, where the hand landmark was detected, but they were outside the detection polygon. The main failure observed in OpenPose was landmark miss rather than detection failure itself, a case where no hands were detected in about 68.3%. For piano applications, no detection is far worse than offset detections. This means that applications relying on live tutoring of piano instruments can filter the offset landmark to the nearest key in the scene, but in case of no detection, no information is captured. For bare

landmark detection in real-time, MediaPipe is proven to be much more useful and resource-efficient than OpenPose.

There is no consistent trend showing detection failure in terms of Fitzpatrick skin tone types. With 16.8% failure rate for Type II, this is the lowest value, while for Type III, 32.8% is the highest. The dataset has a large imbalance in terms of scale data, but the current results suggest MediaPipe does not show any systematic bias against darker palms in the tested range. Type IV and Type V sit in the range of 26.4% and 30.5%. OpenPose's Type II showed a spike of 44.1%, but this result is only equated from 4 participants, which is a minuscule sample size, so the spike might be a noise factor in the raw data rather than a skin tone effect.

For MediaPipe, lighting variation produced effectively zero difference in terms of failure rate. Both Dim and Indoor conditions were 28.6% and 28.7%, respectively. OpenPose showed a slight variation of 21.7% on dim and 29.9% on indoors, this is based on a small detection sample, so it reflects selection bias rather than a real lighting effect. While there is no differentiation between dim and indoor lighting, future research could dive into testing outdoor lighting >500lux or >100lux and even very dark conditions <10 lux illumination.

8.2 MPJPE outcomes

The previous section was focused on detecting whether a fingertip was found during the MIDI signal and video timestamp. This section reports how accurately fingertips were detected for both models, with per-finger results presented using tables and graphs and a breakdown analysis at the end.

8.2.1 MPJPE

As mentioned above in section 6.3.1, only horizontal MPJPE is calculated to account for vertical bias. While OpenPose shows marginally better results for the same test dataset, the detection rate for both models is different. The offline processing of MPJPE and detections for OpenPose took significantly more time than MediaPipe, making MediaPipe more time and resource-efficient. The mean MPJPE values were evaluated to 6.07px for MediaPipe and 5.87px for OpenPose across all videos. Both fall well below the 10-to-11-pixel threshold at the calibrated resolution, meaning both models can identify fingertips when correctly detected. But this applies only to the small subset of detection achieved by OpenPose as discussed in 7.1.6

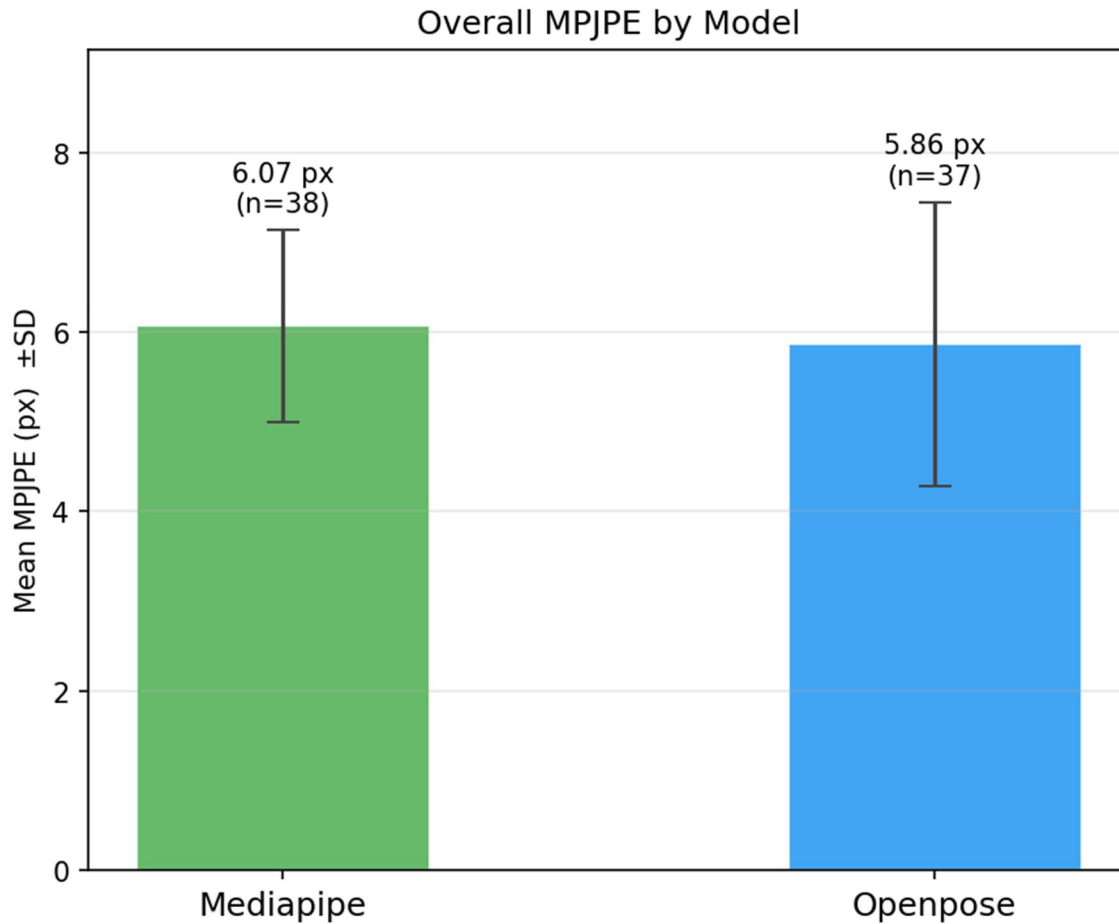


Fig 8.2.1.a: Overall MPJPE on MediaPipe vs OpenPose

Metric	MediaPipe	OpenPose
Mean MPJPE (px)	6.07	5.87
Median (px)	5.86	5.91
SD	1.07	1.59
Min	4.52	2.20
Max	8.74	10.00

Table 8.2.1.b: Overall MPJPE on MediaPipe vs OpenPose Summary

8.2.2 2D Mask Perspective Note

A problem encountered during the mask calibration is accounting for 3d shape of the keys when the camera is angled overhead. Further, the key is from the camera, where more 3D error is introduced with the calibration mask, so most pieces played were positioned near C5, and some notes were rarely at C6. Notes near C6 have introduced minor bias and may contribute to detection miss even if notes were pressed inside the polygon containment box.

Since both models were tested on a polygon mask, the bias is consistent. MediaPipe vs OpenPose comparison remains valid even if the upper register notes are slightly inflated. Further development in this research could replace 2d mask with a 3D system for key boundary or even use specific adjusted 2d masks kept per key for upper keys correction.

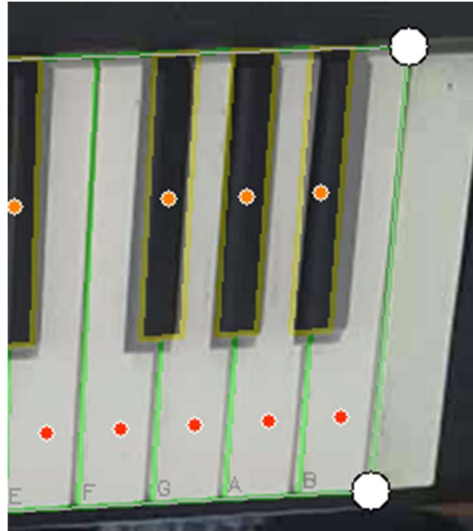


Fig 8.2.2.a: Black keys 2D calibration mask misalignment

8.2.3 MediaPipe Per-Finger MPJPE and Polygon Htrate

Hand	Thumb	Index	Middle	Ring	Pinky
Left	5.7	5.8	6.5	6.6	5.5
Right	5.1	5.7	6.5	6.1	6.0
Max	14.0	10.8	14.7	14.9	12.0
Min	0.0	2.7	1.3	1.9	1.0

Table 8.2.3.a: Per finger MPJPE Summary

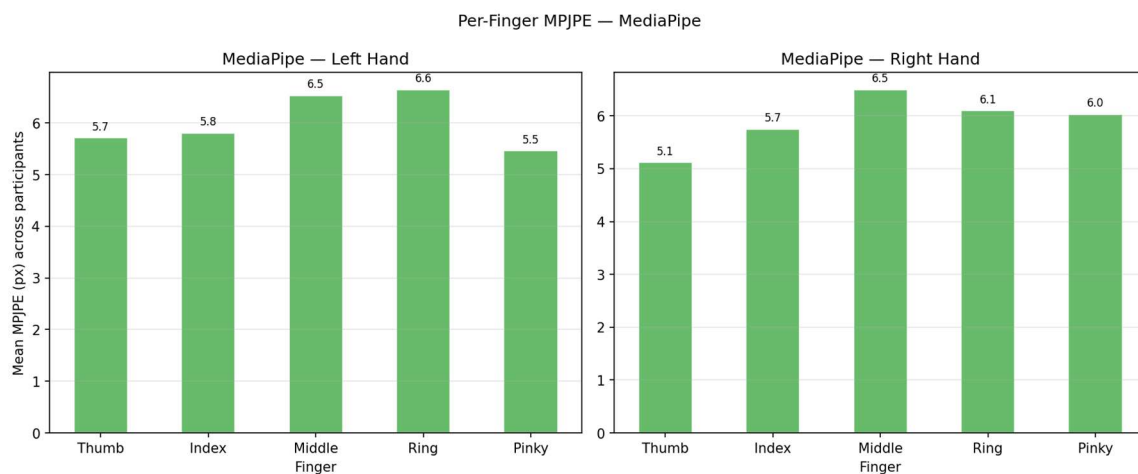


Fig 8.2.3.b: finger breakdown for MPJPE (MediaPipe)

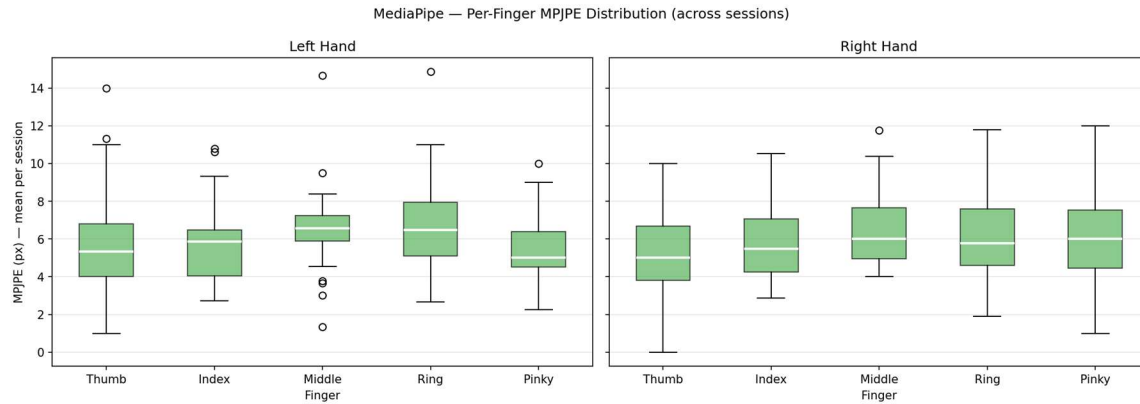


Fig 8.2.3.c: finger breakdown for MPJPE (MediaPipe)

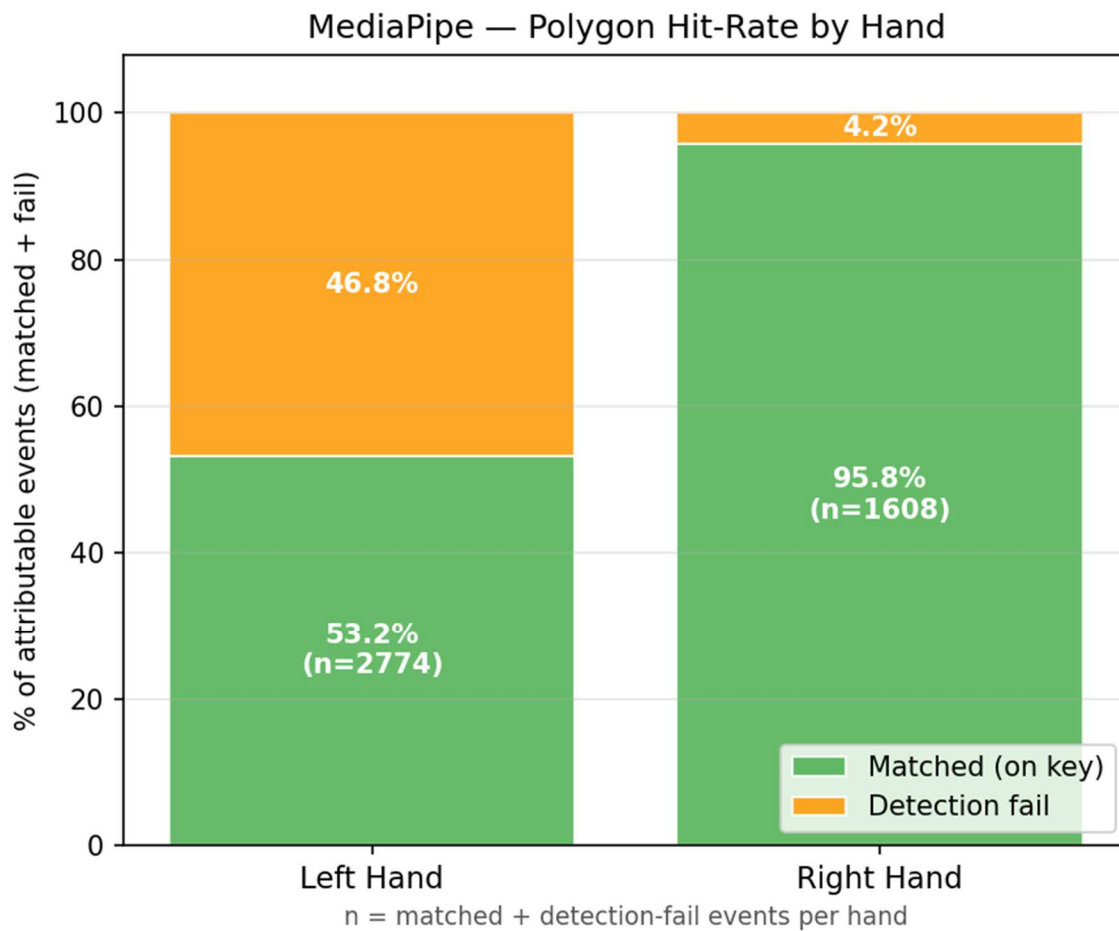


Fig 8.2.3.d: MediaPipe per hand polygon hit rate

8.2.4 OpenPose Per-Finger MPJPE

Hand	Thumb	Index	Middle	Ring	Pinky
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Both	6.9	5.8	6.1	6.1	4.4
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Table 8.2.4.a: OpenPose per finger summary

OpenPose — Per-Finger MPJPE Distribution (across sessions)

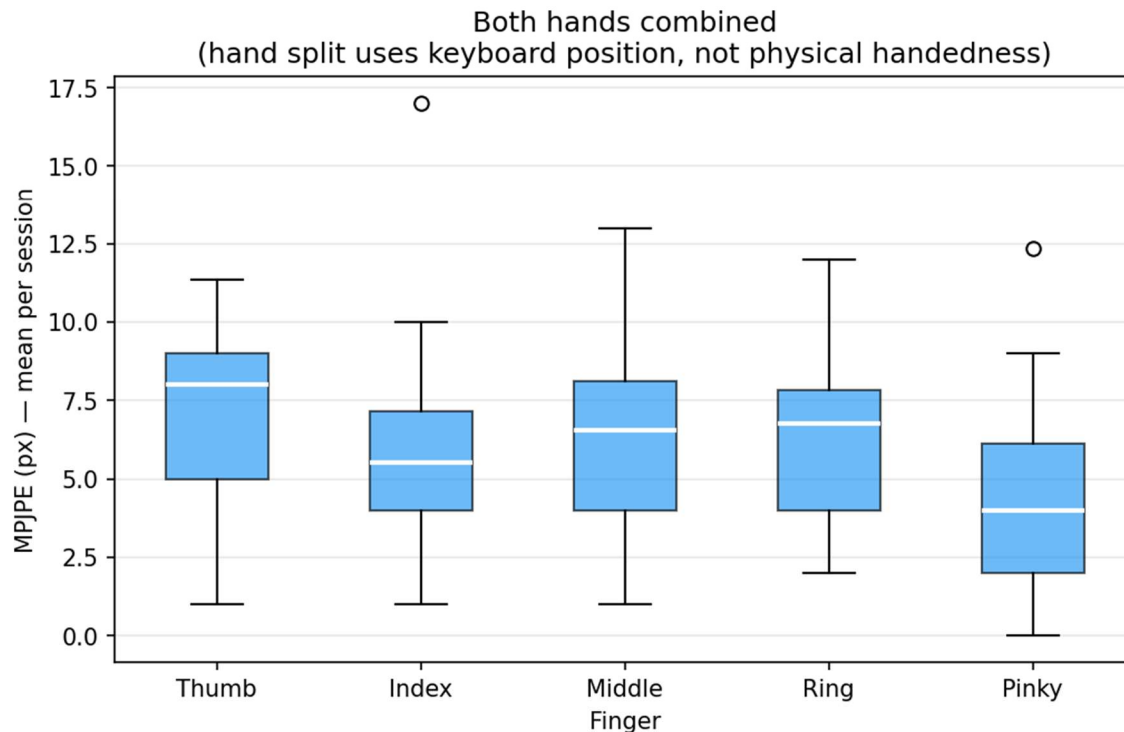


Fig 8.2.4.b: Overall MPJPE combined for both hands (OpenPose)

8.2.5 MPJPE per finger Outcome Breakdown

Across both models, OpenPose showed a slightly lower MPJPE of 5.87px compared to MediaPipe's 6.07px, but this is identified only over 8.1% of events that were detected. The MediaPipe's large sample of 3017 matched events vs 388 for OpenPose should be taken as more representative for accuracy measures. Overall, both models were within the threshold range.

OpenPose shows a different result where the pattern is opposite. The thumb shows the highest MPJPE at 6.9px, while the pinky at 4.4px is the lowest. The two models struggle in different places: MediaPipe, with the middle three fingers anatomically coupled together, and OpenPose, with the most independent finger without tightly cropped preprocessing. Overall, for MPJPE, the OpenPose per finger values are results of a small sample size and should be best interpreted as the model's precision of the given hand region rather than comparing with MediaPipe in this case.

Coming to the Left and Right split for MPJPE, 6.13px for the left and 6.02px for the right, overall is nearly identical. Despite a large asymmetry for detection inside the polygon, i.e., finger landmark detected but outside polygon mask reported in 7.2.3 with graphs where the right hand had 95.8% match rate against 53.2% of left hand, the localization can be identified just as accurately for both. A possible cause for this difference in detection rate between left and right hands might be right-handedness, meaning participants had difficulty playing with their left hands, landing keys inside the key mask properly. Another reason might be MediaPipe's L and R categorization bias, but this prompts separate research to investigate this detection rate difference.

The results for OpenPose were combined for both left and right due to the hand assignments using keyboard position rather than physical handedness, which could misclassify events during two-handed crossing passages. MediaPipe best suits practical deployment on a piano tutoring application due to its reliable detection without any preprocessing, saving processing power. The polygon mask system can be used to assess a player's left and right-hand performance if implemented on a tutor app by assessing the results of section 7.2.3. This makes it possible to run MediaPipe even on decent mobile devices, without requiring any additional hardware cost for a tutoring app.

8.3 Effect of Skin Tone (Fitzpatrick Type) on MPJPE

This section shows the MPJPE results by participants, Fitzpatrick's scale skin type, and explains how each session was classified, and discusses if skin tone has any effect on MPJPE for both models.

8.3.1 Computation of skin tone type

The system of Fitzpatrick scale was developed, which was ITA (individual topology angle) from CIELabs color values sampled across 20 frames to classify into Fitzpatrick groups as mentioned in section 6.2.1, but only manual classification with side-by-side comparison to the Fitzpatrick scale chart in the same section was done. This is because the variation in lighting caused the same participant to be assessed into different types on the scale every time. Manual classification was deemed more accurate than frame estimation.

8.3.2 Fitzpatrick Type and MPJPE

FitzPatrick scale	MediaPipe (MPJPE px)	OpenPose (MPJPE px)
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Type II	5.51	6.41
Type III	5.88	5.30
Type IV	6.39	5.96
Type V	6.29	6.89

Table 8.3.2.a: Fitzpatrick type MPJPE summary

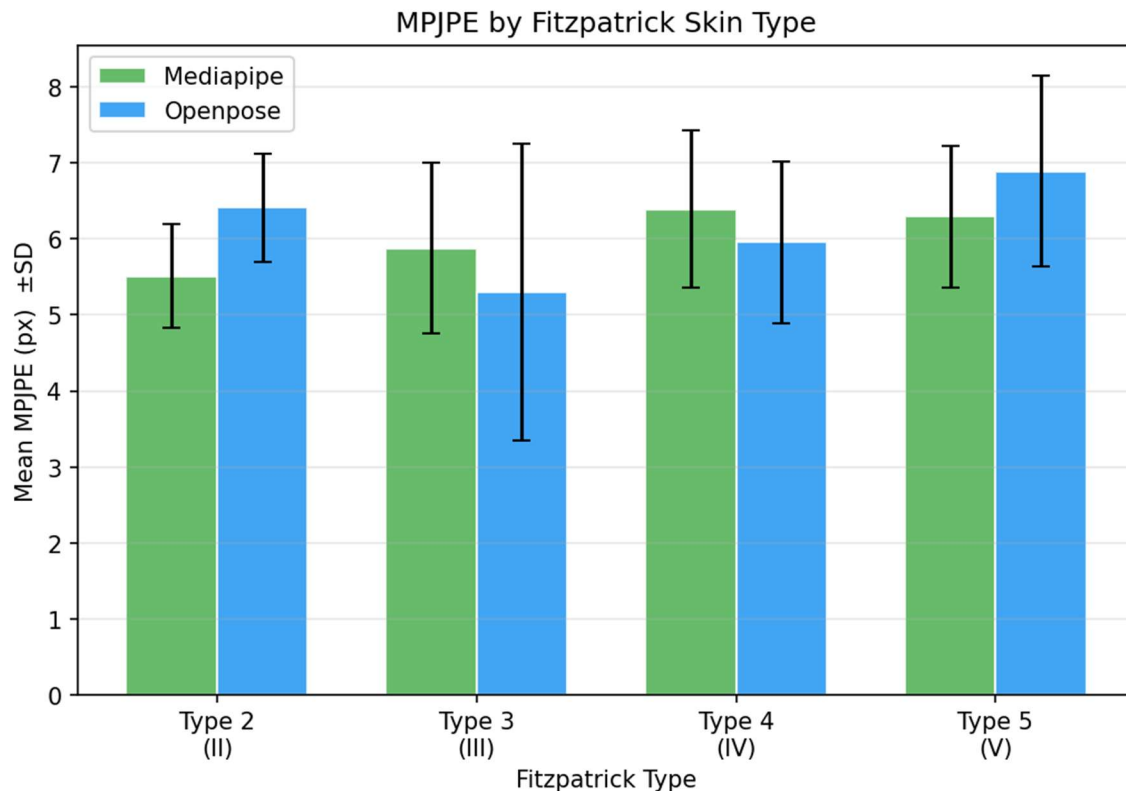


Fig 8.3.2.b: Average MPJPE per Fitzpatrick scale type (II-IV)

8.3.3 MPJPE by Fitzpatrick Breakdown

Looking at the resulting graphs, MediaPipe shows a mild upward trend in MPJPE from lighter to darker types tested. From 5.51 px at Type II to 6.39 px at Type IV, before slightly dropping to 6.29px for Type V. The total range of variation was only 0.88px, which is smaller than the within-group standard deviation of any single group. The trend is not monotonic, and the low variance is shown within the tested types. Hence, MediaPipe doesn't attribute any skin tone bias to the systematic localization from Type II to Type V.

OpenPose shows a different pattern in terms of the FitzPatrick types. Results vary for each type; there is no upward or downward trend, deeming the result inconsistent. While Type II returns to 6.41px, which is the highest, then to 5.30px for Type III, which is the lowest, other types show 5.96 and 6.89px for Type IV and V, respectively. As mentioned in the sections above, OpenPose has a very small sample size (detected landmarks

within the polygon), so this reflects selection bias rather than a real skin tone effect. Hence, OpenPose's results are inconclusive and require a greater sample.

Fitzpatrick Types I and VI were not represented in this study because no participants in these categories volunteered during the recruitment period. Combined with the detection-fail finding in 7.1.6, which showed no consistent direction across types, the conclusion is that MediaPipe demonstrates equitable performance across the Fitzpatrick range tested. Future work should actively recruit participants in Types I and VI with larger and more balanced sample sizes to confirm this finding extends to the extremes of the scale.

Fitzpatrick types I and VI were not represented in this study because no participants in these categories volunteered during the recruitment process. So, MediaPipe demonstrates equitable performance across all types tested. Future work should actively balance the sample data for all types and even include extreme ends of the scale, being type I and type VI

8.4 Effect of Ambient Illuminance on MPJPE

This section opens to the computation of lux and what are the effects of lighting between these models. From lighting categorization to line graphs showing the outcomes of the data sample on MPJPE vs lighting are presented.

8.4.1 Computation of lighting

As mentioned in section 6.2.2, lighting was classified as dim for <100 lux, indoors for >100lux and <1000lux and more than 1000lux as outdoors. The sample dataset doesn't include >1000lux. The illuminance was captured using the light sensor on a mobile phone accessed by a sensor reading app.

8.4.2 MPJPE by Lighting Condition

Lighting condition	MediaPipe	OpenPose
Dim (<100 lux)	6.03	6.17
Indoor (>100 lux)	6.17	5.04
Bright (>1000 lux)	—	—

Table 8.4.2.a: MPJPE by lighting condition Summary

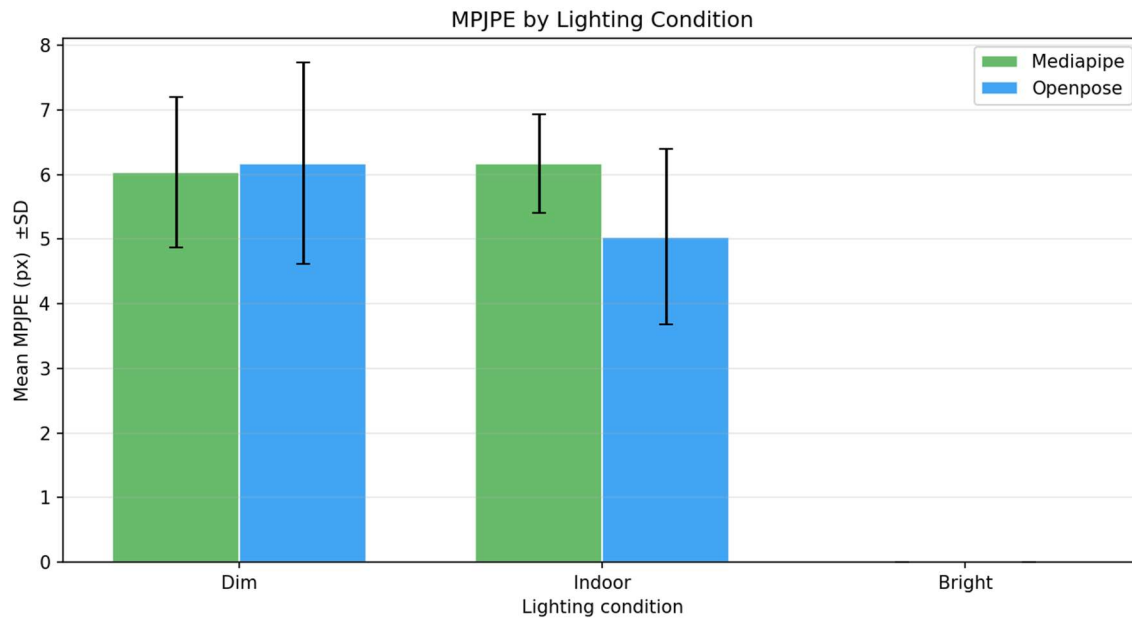


Fig 8.4.2.b: Mean MPJPE per lighting category (no bright conditions were recorded)

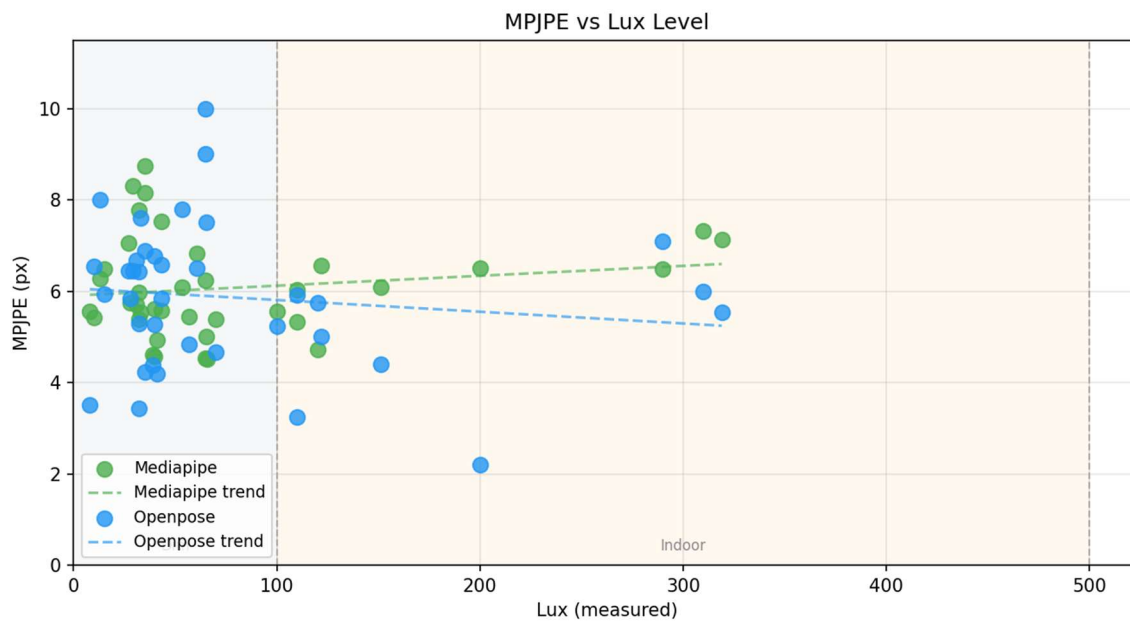


Fig 8.4.2.c: line graph for MPJPE vs Lux levels

Metric	MediaPipe	OpenPose
Slope (px/lux)	0.00217	-0.00257
Intercept (px)	5.90	6.06
Lux range	8.0–319.0	8.0–319.0
n sessions	38	37

Table 8.4.2.d: MPJPE vs Lux Levels Summary

8.4.3 MPJPE by Lighting Breakdown

MediaPipe showed no meaningful difference in the two lighting categories tested. Both conditions returned a similar mean MPJPE of 6.03px (SD = 1.16, n=28) for dim and 6.17 px (SD = 0.76, n = 10) for indoors. The 0.14px difference within the standard deviation of either group suggests MediaPipe's localization accuracy is statistically indistinguishable for the two tested ranges, and lighting doesn't really affect this model's accuracy. The MPJPE vs lux levels graphs support this claim, showing a regression line almost flat across the ranges, and a wide spread of values of MPJPE for Dim bands. This shows that even in dim conditions, a participant can have a good or a bad MPJPE result, so rather than lighting, other factors are contributing to these variations and might heavily depend on individual participants' skills. Hence, lighting levels within the typical home range don't meaningfully affect MediaPipe's tracking quality.

OpenPose has a larger variation between Dim 6.17 px and Indoor 5.04px. While the slope is in the negative meaning, the result was better with brighter lux levels; it only accounts for 27 and 10 detected sessions, respectively, so it should be treated as selection bias, as discussed in previous sections.

Brighter conditions over 500 lux were not represented in this dataset. All sessions had indoor-based lighting or dark lighting, like most piano setups are in. Future work could extend lighting variables to cover both very bright conditions (>500 lux) and very dark (<5lux) extremes to confirm MediaPipe's robustness and even OpenPose with a preprocessing cropping layer.

8.5 Effect of Hand Size on MPJPE

This section goes over the MPJPE effect by hand size and reports if hand size attributes result in more, less, or the same MPJPE values. The results show a scatter plot with regression lines.

8.5.1 Computation of Hand Size

Hand size was manually measured before recording each session using a measuring tape on the hand spread flat from one end of the palm to the other. The measurement was entered on the setup session screen and saved in the session metadata JSON file as mentioned in section 6.2.3. For the analysis, each measurement contributed as one datapoint in the scatter plot, and the results MPJPE of that hand size session were plotted with a regression line using Numpy's least-squares polynomial fit.

8.5.2 MPJPE vs Hand Size

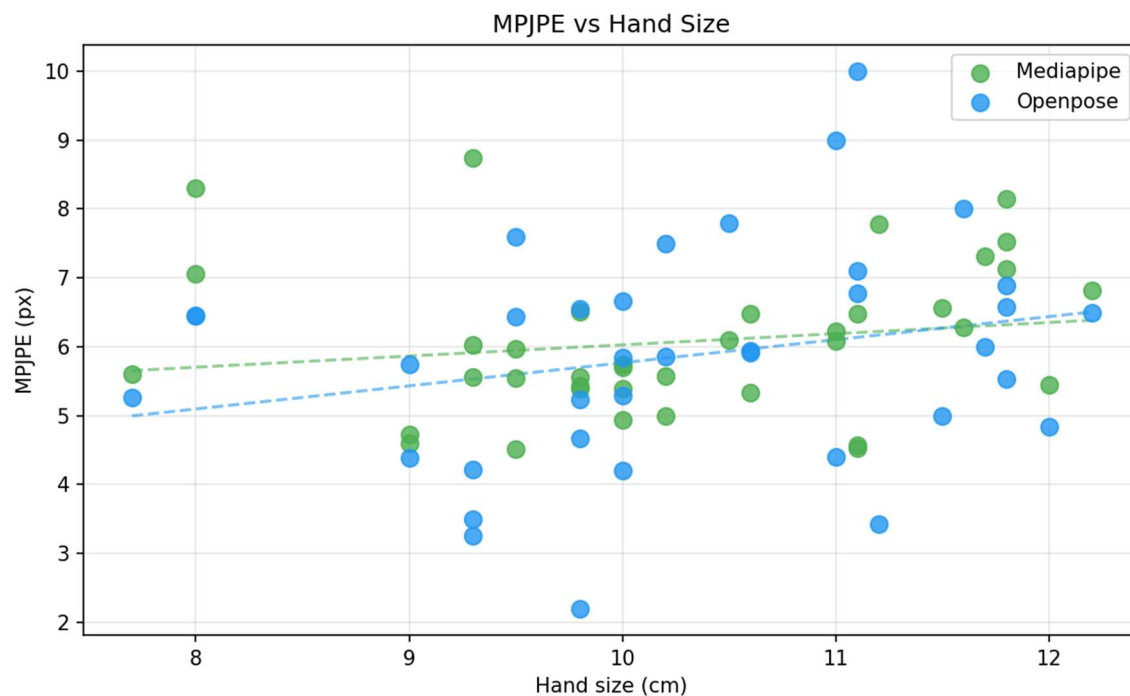


Fig 8.5.2.a: Hand size and MPJPE correlation chart

Metric	MediaPipe	OpenPose
Slope (px/cm)	0.162	0.336
Intercept (px)	4.40	2.41
Hand size range (cm)	7.7–12.2	7.7–12.2
n sessions	38	37

Table 8.5.2.b: Hand size and MPJPE correlation Summary

8.5.3 Hand size effect breakdown

With nearly flat regression lines for both MediaPipe and OpenPose, sizes from 7.7cm to 12.2cm were recorded in this study. MediaPipe's regression slope is 0.162 px per cm with an intercept of 4.40px. OpenPose's slope is 0.336 px per cm with an intercept of 2.41 px. MediaPipe slope is small enough for the full range of hand sizes tested, which only vary in MPJPE by 0.7 px from the smallest to the largest hand. Roughly twice as steep compared to MediaPipe, OpenPose has only 1.5 px variation across the range.

For both regressions, R^2 values are very low at 0.030 for MediaPipe and 0.059 for OpenPose, which indicates hand size explains less than 6% of the variance in the MPJPE for either model. The remaining variance could be attributed to other factors not captured in this analysis, primarily participant skill level and playing style, or even the camera placement variation between sessions.

This implies MediaPipe is reliable across the adult hand size range tested without prior per-use calibration. Both large and smaller hand sizes received comparable accuracy, indicating this variable does not need to be considered when making a consumer-targeted piano tutor app that focuses on adults. The hand sizes did not go below 7.7 cm due to all participants being over 18 years of age. Future work could recruit participants under 18 with guardian consent to test both models' robustness in terms of smaller hand sizes.

9. Summary and Conclusion

9.1 Summary of Project Aim and Outcomes

This research aimed to benchmark vision-based hand tracking models on whether they could reliably identify which finger strikes which key while considering variables like lighting levels, participants' hand size, and participants' skin tone by using MPJPE, the distance from the key's center to the distance between finger landmark dots. Also, this research aimed to give data-backed advice for developing vision-based tutor apps for instruments. Two models were compared to MediaPipe and OpenPose with a MIDI and recorded video synchronized data collection pipeline, where participants played a set of keyboard exercises spanning an octave for white and black keys with two hands. Participants were also required to play finger crossing events. A polygon containment mask was also developed that compared not only the distance from the center of the key but also if the finger tracking model's landmarks were inside the key's border.

The pipeline was used to collect 38 completed sessions, comprising 89560 video frames, 9557 MIDI events, and 4777 note-on events processed. Each session had the lux level measured, participants' hand size, and their Fitzpatrick scale type identified. The test dataset included Fitzpatrick types II to V, hand sizes from 7.7cm to 12.2cm, and lighting levels categorized in dim and indoor conditions.

MediaPipe emerged as the best model in terms of direct deployment on a full scene video. It achieved a mean MPJPE of 6.07px, well below the threshold indicated for the correct key press threshold. MediaPipe also achieved 63.3% in the polygon mask test across all events. OpenPose recorded a marginally lower MPJPE of 5.87px, but only for the low 8.1% events detected successfully due to its working architecture. OpenPose requires pre-cropped hand input preprocessing to properly work. Hence, MediaPipe emerged as the optimal modality to use for a real-time processing use case.

Per hand and per finger results were also tested across all sessions in terms of MPJPE. MediaPipe showed the highest MPJPE for the ring and middle finger, which was consistent with the reduced anatomical independence for these fingers. OpenPose showed the opposite pattern, with the thumb being the weakest and the pinky being the most accurate. A substantial poly hit rate difference for MediaPipe's left and right was observed, equating 95.8% for the right hand and 53.2% for the left hand for both hands

playing the identical key. The asymmetry could be cause of righthandedness and participants finding the left hand more difficult to play with, rather than MediaPipe's hand bias. This is for further investigation and research in future work.

Skin tone, ambient lighting, and hand size all showed negligible effects on MPJPE and detection in the first place for MediaPipe, but for OpenPose, due to the absence of preprocessing of cropped hand data, the result is inconclusive and remains open for further investigation. This showed MediaPipe is viable for tutoring app deployment.

9.2 Novel Contributions

Several outcomes extend beyond what exists in prior piano finger tracking literature.

A dataset of 38 piano performances with MIDI notes on events, temporal frame alignment, and multiple classification variables like Fitzpatrick skin type, hand size in centimeters, and lux levels was produced. This dataset contained standardized playing patterns repeated by every participant, enabling cross-checking and participant comparison on identical inputs. The closest dataset resources separate these elements: MAESTRO (Hawthorne et al., 2019) and MAPS (Emiya, Badeau, and David, 2010) provide piano audio and MIDI, while general hand poses datasets such as FreiHAND (Zimmermann et al., 2019) provide annotated keypoints without MIDI or piano context. The dataset produced by this research fills this gap by providing output of the video with MIDI timings and with the variables mentioned above.

Second is a new acquisition method to align MIDI note events with video frames. This makes sure that MIDI events and video data share a common time anchored at recording, which avoids the issue of misalignment during analysis. A single call of `time.pref_counter()` fixes this issue. This sub-frame temporal alignment also enables ground truth annotation for fingers without manually labeling each frame of the video.

Third, a homography-calibrated 2D polygon mask was developed to test if the finger landmarks of both models were in the same 2D space as the key's boundaries. The sizable box was computed with Siddique et al's. (2019) paper on perspective homography calibration for warping quadrilaterals on code and on screen. This system was developed to counter oblique camera angles more reliably while isolating horizontal localization errors. This system now allows bias reduction for camera angles that are not straight 90 degrees, or make any application relying on instrument analysis measure only a part of the piano due to its sizable dimensions and number of key variables.

Fourth, the MPJPE metric only measured horizontal axis error. This is a methodological contribution to account for the full Euclidean 2D error reported in existing studies such as Liang et al. (2016), Hiranaka et al. (2022), and Mueller et al. (2018), which conflate horizontal mispositioning (the dimension that determines key identification) with vertical displacement caused by finger length and camera depth angle. The problem of horizontal

mispositioning with vertical displacement caused by the finger length and the camera depth angle is handled in this case. MPJPE is isolated for the dimension that matters for piano key identification and essentially makes the entire vertical distance in the middle act as a center.

9.3 Room for Future Development

Finding participants who fell on the type I and VI of the Fitzpatrick scale was not possible in this project's timeline. Future expansion in this research or similar work can aim to balance underrepresented participants and, overall, a balanced sample across the full range of the scale. Without two extreme ends of the scale, performance and claims for MediaPipe cannot be made, hence only the middle spectrum results are shown.

More participants could be involved in capturing more than 40 videos. Also, participants below the age of 18 could be included with parental consent, which could help to search for a pattern in hand size data. This could test if MediaPipe still generalizes to smaller hands. From 7.5-12.5cm was only captured in this dataset covering adults.

Different filtering options, like Kalman filtering or optical flow-based interpolation, could be developed to account for finger overlapping during fast keystrokes. These filter options could allow direct reduction of the detection failure rate of MediaPipe, which was observed to be 28.7% from the entire dataset. Also, research on the left- and right-hand polygon mask accuracy could be explored to know the exact reason for the high left-hand error on MediaPipe.

The dataset was only collected among inexperienced piano players. Experienced players with different complex musical pieces could be tested for more rigorous movement data collection. This research only required the participants to play basic notes and patterns, which could be learned in a few minutes. Cross-checking MIDI events, checking where the most detection failures occurred, or where the landmark check failed could also help identify whether the model fails around specific tempos, hand positions, or finger combinations.

The data captured was entirely done indoors (less than 100lux) and in dim conditions (more than 100 lux, less than 1000 lux). Future work could expand into more complex lighting ranges like super dim, dim, indoors, daylight, or even different types of light bulbs used. Testing this can confirm if any problems with lighting robustness exist within the model.

For OpenPose, a preprocessing layer to tightly crop hand images before passing it to the model's network can be developed. This could prompt a fairer comparison between MediaPipe and OpenPose.

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11. Appendices

A. Initial Plan Document

Level 6 Production Project	
Project Initiation	
Name: Aayam Regmi	Student I.D.: 77466864
Course: BSc (Hons) Computing	Supervisor's Name: Mr. Saroj Sharma
Final Project Individual Aim & Objectives	
Title of my Project: Comparative Evaluation of Vision-Based Finger Tracking for Piano Interaction	
Aim of my Project: To evaluate robustness, accuracy and real-time performance of two or more open-source vision-based finger tracking systems during interaction with a real piano acting as a ground truth. Considering lighting conditions, finger and hand occlusion challenges, and user diversity (hand size and skin tone) the aim is to determine whether these variables affect the consistency, performance and accuracy across different pretrained models.	
Objectives of my Project: • To develop a fundamental understanding of vision-based hand and finger tracking algorithm techniques including temporal tracking, landmark detection and handling occlusions between overlapping hands and fingers. • To analyze existing computer vision finger tracking approaches in real-time. • To design a system with a camera programmed to work with finger tracking and a real electronic piano for ground truth interaction. • To establish a camera-to-keyboard calibration system that maps pixel coordinates to physical key positions. • To compare multiple pretrained finger-tracking models (MediaPipe Hands, OpenPose, MoveNet) under same experimental conditions and evaluate the result. • To define and implement quantitative performance metrics for finger tracking evaluation. • To analyze different lighting situations and its effect on tracking accuracy and stability. • To consider tracking performance across users with different hand sizes. • To assess performance bias when using tests with broad skin tone categories. • To investigate the impact of finger and hand occlusion on tracking accuracy and recovery behavior.	

- To evaluate trade-offs between accuracy and real-time performance between different pretrained models and tabulate the results.
- To identify current limitations of approaches in vision tracking models in piano interaction contexts and recommend targeted improvements.
- To establish a reproducible experimental protocol for evaluating hand-tracking systems in the context of music and instruments.

Specification of my Product:

Functional Requirements

#	Functional Requirement	Description / Scope	MoSCoW
1	Finger landmark tracking using vision models	Using preexisting finger tracking models with a fixed camera above a real piano	Must
2	MIDI ground truth capture	Establish ground truth with digital MIDI output of Casio CTK-2400	Must
3	Temporal synchronization & comparison	Tabulate comparison between different models, fingertip position and keypress times	Must
4	Support multiple open-source models	Integrate & compare at least MediaPipe Hands, OpenPose, MoveNet (Thunder)	Must
5	Camera-to-keyboard spatial calibration	Map pixels to physical piano or establish virtual landmarks	Must
6	Data logging system	Store x,y coordinates, video frames and timestamps provided by MIDI	Must
7	Keypress event analysis	Calculate MIDI keypress and fingertip Euclidean distance, marking active finger or failure	Must
8	Quantitative performance metrics calculation	Calculate primary metric MPIPE and secondary metrics like loss rate, temporal jitter, occlusion recovery time, FPS latency	Must
9	Controlled variation of experimental conditions	Test different scenarios like hand size, skin tones and occlusion	Must
10	Statistical comparison of models	Perform statistical tests (e.g. t-test) to compare models across conditions	Must
11	Model selection justification	For piano interaction provide scientific reasoning and recommendations for the best suitable model	Must
12	Reproducible experimental protocol	Record a detailed protocol for future enhancements and recreations	Should

13	Evaluation under additional models	Include one or more extra models beyond the main three (e.g. BlazePose, HRNet, custom fine-tuned, etc.)	Could
14	Real-time visualization	Add detection landmark and active finger indicator live overlay interface during data collection	Should
15	Automatic detection & annotation of occlusion events	For severe occlusion cases automatically classify and log frames	Could
16	3D hand pose estimation	Use models that output 3D coordinates with multi camera system	Won't
17	Mobile / embedded deployment	Deploy the final comparison system on mobile devices or edge hardware	Won't
18	Multi-camera / multi-view tracking	Use more than one camera for improved accuracy	Won't
19	Real-time audio feedback during playing	Generate sound or visual feedback based (gamification)	Won't

Non-Functional Requirements			
#	Non-Functional Requirement	Description / Expectation	MoSCoW
1	Research-focused with functional prototype	Deliver meaningful research insights supported by a working prototype system	Must
2	Fair & controlled comparison conditions	models evaluated under identical hardware, lighting, camera position & setup	Must
3	Development using Python & established open-source libraries	Use Python + OpenCV, MediaPipe, MIDI libraries, stats & plotting as primary tools	Must
4	Ethical compliance & participant protection	Follow university ethics guidelines, obtain approval, informed consent, anonymized & secure data storage	Must
5	Reasonable real-time performance	Achieve at least 15–20 FPS on the development machine during data recording	Should
6	Good code quality, documentation & reproducibility	Clean code structure and comprehensive README	Should
7	Basic robustness & graceful error handling	System logs errors and continues operation (no full crash)	Could

Research:

Finger tracking systems are used in many fields today, including human-computer interaction (HCI), virtual and augmented reality, and medical assessments [1]. In modern frameworks for finger tracking, accuracy is the deciding factor for usability, and often research done under such libraries have controlled datasets that might not account for multiple real-world interaction challenges [2]. From lighting variations to finger occlusion and user diversity, a piano has many of these challenges that can test these frameworks' accuracy [3].

Due to vision-based tracking systems relying on RGB image information, skin tone is included in this project as an evaluation factor [4]. Multi-range RGB values introduce different contrasts and reflections under different lighting conditions, which can influence landmark detection evaluation [5]. Studies conducted under vision systems can exhibit disparities in performance across demographic groups [6]. Multiple pretrained models might be trained under biased RGB values with limited demographic diversity [7]. With broad skin tone assessments, this study will be able to assess potential bias in the system, determining potential dataset improvements for vision tracking models [4][7].

Hand size is also considered, as differences in finger length and joint spacing may affect landmark localization accuracy [2]. Piano interaction presents unique challenges, including rapid finger movements, frequent hand occlusion, and proximity of fingers, making it an ideal testbed for tracking system robustness [3][8]. Additionally, multi-model tests can determine if certain behaviors are model-specific or a generalized problem in terms of finger tracking systems [2][9].

Evaluation:

The average Euclidean distance between predicted fingertip locations by the open-source models and the corresponding pressed key location is the Mean Per-Joint Position Error (MPJPE) [10]. MPJPE will be the primary performance metric for this research. If MPJPE is a negligible value like <10mm, the tracking will be considered accurate [10][11].

Secondary metrics:

- Loss rate (percentage of frames where hands are not detected)
- Temporal jitter (standard deviation of joint positions across continuous frames)
- Occlusion recovery time (frames required to re-establish tracking)
- Latency (FPS) [2][9][12]

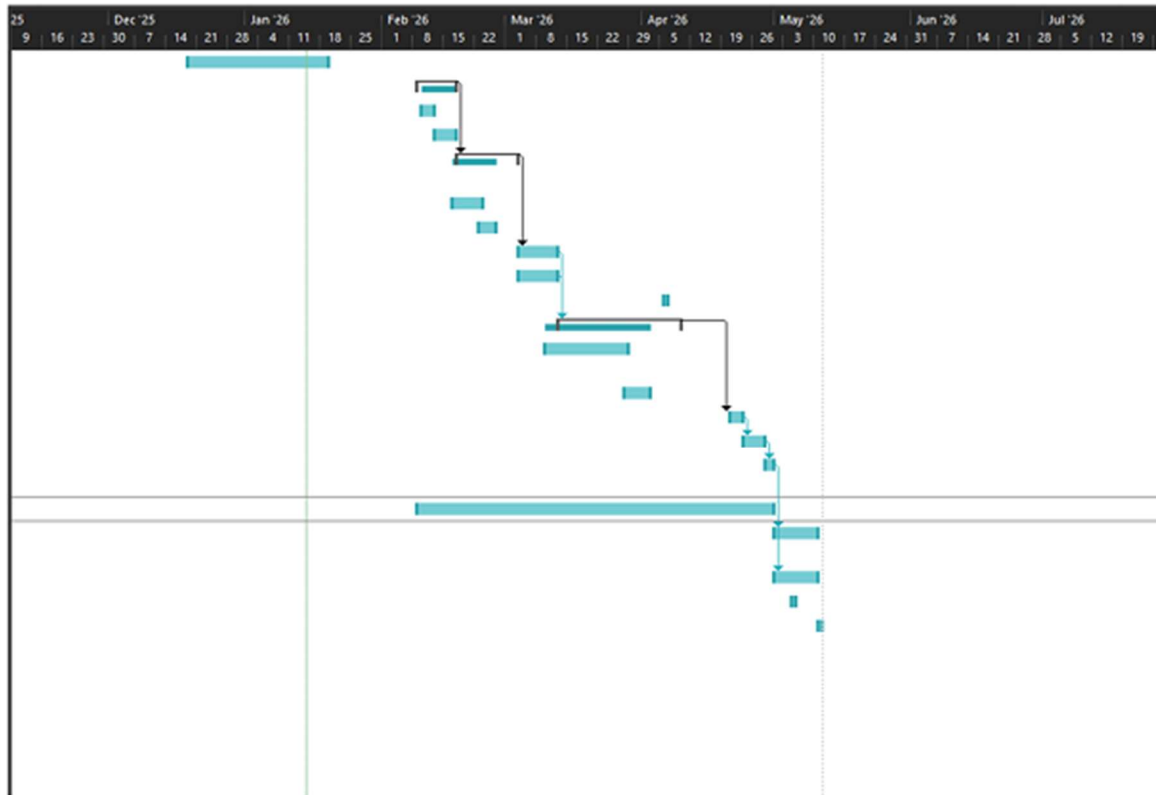
The dataset and results will be categorized by the following standards:

1. Lighting categorization:
 - standard indoor 400-500 lux
 - bright 800-1000 lux
 - dim 150-200 lux [2]
2. Hand size categorization: Small · Medium · Large [2]

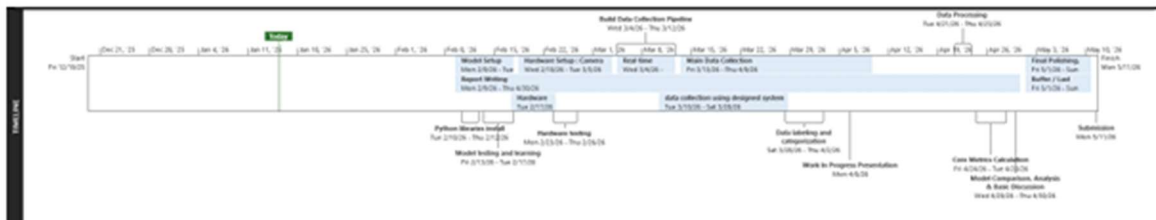
3. Skin tone category method: Fitzpatrick Scale [4][13] Statistical analysis will be paired with a t-test ($p < 0.05$) to compare model performance [2][9]. Validation of ground truth generation will be done by annotating 10% frames [11].
Project Planning & Methodology
Project Planning: Task Sheet

	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1		Project initiation + risk register	22 days	Fri 12/19/25	Mon 1/19/26		
2		Model Setup and Testing	7 days	Mon 2/9/26	Tue 2/17/26		
3		Python libraries install	3 days	Tue 2/10/26	Thu 2/12/26		
4		Model testing and learning	3 days	Fri 2/13/26	Tue 2/17/26		
5		Hardware Setup : Camera and Piano Calibration	10 days	Wed 2/18/26	Tue 3/3/26	2	
6		Hardware purchase	5 days	Tue 2/17/26	Mon 2/23/26		
7		Hardware testing	4 days	Mon 2/23/26	Thu 2/26/26		
8		Build Data Collection Pipeline	7 days	Wed 3/4/26	Thu 3/12/26	5	
9		Real-time Overlay System Creation	7 days	Wed 3/4/26	Thu 3/12/26		
10		Work In Progress Presentation	1 day	Mon 4/6/26	Mon 4/6/26		
11		Main Data Collection	20 days	Fri 3/13/26	Thu 4/9/26	8,9	
12		data collection using designed system	15 days	Tue 3/10/26	Sat 3/28/26		
13		Data labeling and categorization	5 days	Sat 3/28/26	Thu 4/2/26		
14		Data Processing	3 days	Tue 4/21/26	Thu 4/23/26	11	
15		Core Metrics Calculation	3 days	Fri 4/24/26	Tue 4/28/26	14	
16		Model Comparison, Analysis & Basic Discussion	2 days	Wed 4/29/26	Thu 4/30/26	15	
17		Report Writing	59 days	Mon 2/9/26	Thu 4/30/26		
18		Final Polishing, Proofreading, Formatting	7 days	Fri 5/1/26	Sun 5/10/26	16	
19		Buffer / Last-minute Fixes	7 days	Fri 5/1/26	Sun 5/10/26	16	
20		Research Presentation	1 day	Tue 5/5/26	Tue 5/5/26		
21		Submission	1 day	Mon 5/11/26	Mon 5/11/26		

Gantt chart



Project Timeline



Methodology:

This project adopts a Rapid Application Development (RAD) methodology because the data collection and processing will need higher priority and time after the system is developed.[14] Other methodologies such as waterfall emphasizes a straightforward development with no turnback's, while RAD prioritizes rapid development and

iterations of a working prototype. For comparative research projects like this, RAD is appropriate as core functional testing platform is required rather than product ready application. RAD allows continuous feedback and changes from the supervisor's advice as well.

Resources

The software I require to complete my Project successfully:

Item	Source
Windows OS	Own
Python 3.8+	Own
MediaPipe	Open Source
OpenCV	Open Source
MIDI Interface Libraries (mido, python-rtmidi)	Open Source
Statistical Analysis Software (scipy, statsmodels)	Open Source
Data Visualization (Matplotlib, Seaborn)	Open Source
Version Control (Git, GitHub)	Open Source
Microsoft Word	Own
Microsoft Excel	Own
Microsoft PowerPoint	Own
Web browsers: Google Chrome	Own

The hardware I require to complete my Project successfully:

Item	Source
Laptop / PC (16GB RAM, i7-14 th gen CPU, RTX 5070)	Own
USB Camera (30 FPS, 1080p resolution)	Own
Casio CTK-2400 Digital Piano	Own
USB A-to-B Cable (for MIDI connection)	Purchase
Camera Tripod or Mount	Purchase
Digital Lux Meter (for lighting measurement)	Purchase
Colored Calibration Markers	Purchase

Human Resource	
I am working on my Project with the following people	
Name: Aayam Regmi	Role:
	Module Leader: Dr. Rohit Raj Pandey Supervisor: Mr. Saroj Sharma
Initial Bibliography	
1. Zhang, F., et al. (2020). MediaPipe Hands: On-device Real-time Hand Tracking. arXiv preprint arXiv:2006.10214. 2. Jo, H., & Kim, S. (2022). Comparative Analysis of OpenPose, PoseNet, and MoveNet Models for Pose Estimation in Mobile Devices. Traitement du Signal. 3. Liang, H., et al. (2016). Barehanded Music: Real-time Hand Interaction for Virtual Piano. ACM Symposium on Interactive 3D Graphics and Games. 4. Buolamwini, J., & Gebru, T. (2018). Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. Proceedings of Machine Learning Research. 5. Groh, M., et al. (2023). Robustness Evaluation in Hand Pose Estimation Models using Metamorphic Testing. arXiv preprint. 6. Mehrabi, N., et al. (2021). A Survey on Bias and Fairness in Machine Learning. ACM Computing Surveys. 7. Cho, S., et al. (2023). Consensus and Subjectivity of Skin Tone Annotation for ML Fairness. NeurIPS. 8. Hiranaka, A. (Year). AR Piano Playing Using Real-Time Hand Tracking. 9. Kim, et al. (2022). Comparative Analysis of Skeleton-Based Human Pose Estimation. MDPI Future Internet. 10. Mueller, F., et al. (2018). GANerated Hands for Real-Time 3D Hand Tracking from Monocular RGB. CVPR. 11. Salz, D., & Azam, F. (2019). Playing a Virtual Piano with Dynamics. Stanford University Report. 12. Cao, Z., et al. (2019). OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields. IEEE TPAMI. 13. Fitzpatrick, T.B. (1988). The Validity and Practicality of Sun-Reactive Skin Types I Through VI. Archives of Dermatology. 14. Martin, J. (1991). Rapid Application Development. Macmillan.	

B. Risk Register Document

Level 6 Production Project	
Risk Register	
Name: Aayam Regmi	Student I.D.: 77466864
Course: BSc (Hons) Computing	Supervisor's Name: Mr. Saroj Sharma
Title of my Project: Comparative Evaluation of Vision-Based Finger Tracking for Piano Interaction	
Severity calculation: (Likelihood) x (Impact)	
- Low: 1-5 - Medium: 6-12 - High: 13-25	

ID	Risk	Risk Description	Likelihood	Impact	Severity Level	Owner	Mitigation	Status
1	Inaccurate or unstable finger tracking due to occlusion	Frequent occlusion due to fast hand movements while playing the piano	Medium (4)	Medium (3)	(12) Medium	Aayam Regmi	Have datasets with controlled tempos and occlusion levels	Closed
2	Poor model performance under varying lighting conditions	Inconsistency due to light level changes	Medium (3)	Medium (3)	(9) Medium	Aayam Regmi	Conduct all data collection under controlled lighting environment (dim/standard/bright)	Closed
3	Bias or degraded performance across skin tones and hand sizes	Performance disparity in training data categorized by Fitzpatrick scale	Medium (3)	Medium (3)	(9) Medium	Aayam Regmi	Have diversity in participant recruiting in hand sizes and skin tones and perform statistical tests on Fitzpatrick scale groups	Closed
4	Synchronization issues between video & MIDI ground truth	Unsynchronized comparison due to timestamp mismatch	Low (2)	High (5)	(10) Medium	Aayam Regmi	Use precision timestamps and validate test data using annotations	Closed
5	Real-time performance below target (FPS < 15-20)	Low model performance on hardware	Low (2)	Low (2)	(4) Low	Aayam Regmi	Use lighter models and use NVIDIA CUDA acceleration supported models or just multi core CPU procession	Closed
6	Ethical approval or participant consent delays	Delay in approval of research or issues with data	Medium (3)	High (5)	(15) High	Aayam Regmi	Prepare ethical consent form and	Closed

		consent and protection					discuss with supervisor regularly	
7	Hardware failure, malfunction or unavailability	Hardware failure due to various reasons like static charge, water damage or unavailability	Low (2)	High (5)	(10) Medium	Aayam Regmi	Test hardware before testing, if hardware gives out during testing purchase backup hardware, borrow backup piano for project duration in case of failure	Closed
8	Time overrun due to integration & debugging complexity	Not being able to meet MS projects deadlines due to implementation and data collection	Medium (3)	High (5)	(15) High	Aayam Regmi	Follow RAD methodology and complete testing system before MS project deadline, Prioritize more time for data collection	Closed
9	Data loss or corruption during long recording sessions	Video datasets lost due to software crashes or memory corruption	Low (1)	Medium (3)	(3) Medium	Aayam Regmi	Have backup data in GitHub, rerecord data incase backup does not exist	Closed

Summary:

Likelihood:

- 4 low
- 5 medium

Impact:

- 1 low
- 4 medium
- 4 high

Severity:

- 1 low
- 6 medium
- 2 high

C. Ethics Approval

STAGE 1 - RESEARCH ETHICS APPROVAL FORM (from December 2016)



STAGE 1 - RESEARCH ETHICS APPROVAL FORM

Research by students and staff at the University must receive ethical approval before any data collection commences. Applications may be made on the Research Ethics Online system or via approval forms.

If using the approval forms, applicants complete this [Stage 1 - Research Ethics Approval Form](#) which includes the Risk Checklist.

For student projects classified as Risk Category 1 (e.g., many literature reviews), these can be approved on this [Stage 1 - Research Ethics Approval Form](#) by the Research Supervisor.

Applicants whose research studies are classified as Risk Category 2 or 3 must also complete and submit the separate [Stage 2 - Research Ethics Approval Form](#).

Guidance for completion of this form and the application process is provided on pages 3 and 4.

APPLICANT DETAILS	
Your name (if a group project, include all names)	Aayam Regmi
School	The British College
STATUS	
• Undergraduate student	☒
• Taught Postgraduate student	☐
• Research Postgraduate student	☐
• Staff member	☐
• Other (give details)	
IF THIS IS A STUDENT PROJECT	
• Student ID	77566864
• Course title (eg. BA (Hons) History)	Bsc (Hons) Computing
• Student email	raayam22@tbc.edu.np / A.Regmi2995@student.leedsbeckett.ac.uk
• Research Supervisor's name Or Director of Studies' name	Mr. Saroj Sharma
THE PROJECT/STUDY	
Project /study title	Comparative Evaluation of Vision-Based Finger Tracking for Piano Interaction
Start date of project	19 th December, 2025
Expected completion date of project	5 th May, 2026
Project summary – please give a brief summary of your study (maximum 100 words)	
To evaluate robustness, accuracy and real-time performance of two or more open-source vision-based finger tracking systems during interaction with a real piano acting as a ground truth. Considering lighting conditions, finger and hand occlusion challenges, and user diversity (hand size and skin tone) the aim is to determine whether these variables affect the consistency, performance and accuracy across different pretrained models.	
CONFIRMATION STATEMENTS	
The results of research should benefit society directly or by generally improving knowledge and understanding. Please tick this box to confirm that your research study has a potential benefit. If you cannot identify a benefit you must discuss your project with your Research Supervisor to help identify one or adapt your proposal so the study will have an identifiable benefit.	☒
Please tick this box to confirm you have read the Research Ethics Policy and the relevant sections of the Research Ethics Procedures and will adhere to these in the conduct of this project.	☒

RISK CHECKLIST - Please answer ALL the questions in each of the sections below – tick YES or NO		YES	NO
WILL YOUR RESEARCH STUDY?			
1	Involve direct and/or indirect contact with human participants?	☒	☐
2	Involve analysis of pre-existing data which contains personal or sensitive information not in the public domain?	☐	☒
3	Require permission or consent to conduct?	☒	☐
4	Require permission or consent to publish?	☒	☐
5	Have a risk of compromising confidentiality?	☐	☒
6	Have a risk of compromising anonymity?	☐	☒
7	Collect / contain sensitive personal data?	☐	☒
8	Contain elements which you OR your supervisor are NOT trained to conduct?	☐	☒
9	Use any information OTHER than that which is freely available in the public domain?	☒	☐
10	Involve respondents to the internet or other visual/vocal methods where participants may be identified?	☐	☒
11	Include a financial incentive to participate in the research?	☐	☒
12	Involve your own students, colleagues or employees?	☒	☐
13	Take place outside of the country where you are enrolled as a student, or for staff, outside of the UK?	☒	☐
14	Involve participants who are particularly vulnerable or at risk?	☐	☒
15	Involve any participants who are unable to give informed consent?	☐	☒
16	Involve data collection taking place BEFORE informed consent is given?	☐	☒
17	Involve any deliberate deception or covert data collection?	☐	☒
18	Involve a risk to the researcher or participants beyond that experienced in everyday life?	☐	☒
19	Cause (or could cause) physical or psychological harm or negative consequences?	☐	☒
20	Use intrusive or invasive procedures?	☐	☒
21	Involve a clinical trial?	☐	☒
22	Involve the possibility of incidental findings related to health status?	☐	☒
23	Fit into any of the following security-sensitive categories: concerns terrorist or extreme groups; commissioned by the military; commissioned under an EU security call; involves the acquisition of security clearances? If yes, see the guidance.	☐	☒

CLASSIFICATION	Tick the box which applies to your project
The following guidance will help classify the risk level of your study	
If you answered NO to all the above questions, your study is provisionally classified as Risk Category 1 (literature reviews will be Risk Category 1).	☐
If you answered YES to any question from 1-13 and NO to all questions 14-22, your study is provisionally classified as Risk Category 2 .	☒
If you answered YES to any question from 14-22, your study is provisionally classified as Risk Category 3 .	☐
If question 23 has been answered YES , your application will be reviewed by the Chair of the University Research Ethics Sub-committee	☐

DECLARATION AND SIGNATURE/S			
I confirm that I will undertake this project as detailed above. I understand that I must abide by the terms of the approval and that I may not make any substantial amendments to the project without further approval.			
Signed		Date	3 rd March, 2026

FOR RISK CATEGORY 1 STUDENT PROJECTS			
Approval from the Research Supervisor or Director of Studies for a student project: I have discussed the ethical issues arising from the project with the student. I approve this project.			
Name		Signed	
		Date	

NEXT STEP
RISK CATEGORY 1 PROJECTS: IF YOUR PROJECT HAS BEEN CLASSIFIED AS RISK CATEGORY 1: - Students: The Research Supervisor should return the signed form to the student and send a copy to the Local Research Ethics Co-ordinator and where relevant, the Research Module Leader, for information. - Staff: Submit this form to your Local Research Ethics Co-ordinator. RISK CATEGORY 2 OR 3 PROJECTS: IF YOUR PROJECT HAS BEEN CLASSIFIED AS RISK CATEGORY 2 OR 3 please complete the <u>Stage 2 - Research Ethics Approval form</u> and submit both forms together with supporting documentation. QUESTION 23: If this question has been answered YES, your application will be reviewed by the Chair of the University Research Ethics Sub-committee, and the forms should be submitted directly to Professor Karl Spracklen, k.spracklen@leedsbeckett.ac.uk. You will need to submit the Security-sensitive research form available from the Research Ethics web page. <i>Research ethics application forms will be retained in the School for the purposes of quality assurance of compliance and audit for THREE years</i>

NOTES FOR COMPLETION

University Research Ethics Policy and Procedures: The University Research Ethics Policy and Research Ethics Procedures should be read prior to commencing this application. Consideration of the application by the reviewer/s will be undertaken in accordance with the Policy and Procedures.

External requirements for the project: Applicants should consider if there are requirements by any relevant professional, statutory or regulatory body, or learned society, which may be relevant to the project or if the project also requires external approval.

Submission

- Student applicants: email the typed form/s to your Research Supervisor or Director of Studies.
- Staff applicants: email the typed form/s to your Local Research Ethics Co-ordinator.

How to complete the form

You can navigate through the form by using the tab keys. If you prefer to complete a normal Word document, you can unlock the form by selecting the 'Restrict Editing' button on the Developer tab, then click on 'Stop Protection'. The boxes should expand to allow space for your text.

Signatures

Electronic/typed signatures are acceptable for emailed forms, as the emails provide the audit trail for all parties' agreement and approval of the forms (e.g., student applicant → Research Supervisor → Local Research Ethics Co-ordinator).

Outcome

Applicants will be advised of the outcome of the application by receipt of the signed form from:

- The Research Supervisor or Director of Studies for Risk Category 1 student projects;
- The Local Research Ethics Co-ordinator or the School level group for Risk Category 2 and 3 projects.

YOU MAY ONLY BEGIN ANY DATA COLLECTION ONCE YOU RECEIVE NOTIFICATION THAT THE PROJECT HAS ETHICAL APPROVAL. If the circumstances of your research study change after approval it is your responsibility to revisit the Risk Checklist and complete a further application.

Advice

When completing the Stage 1 - Research Ethics Approval Form, if you are uncertain about the answer to any question, read the relevant section of the Research Ethics Procedures document, and if you are still unsure:

- if you are student, seek guidance from your Research Supervisor or Director of Studies;
- if you are a staff member, contact your Local Research Ethics Co-ordinator.

APPROVAL PROCESS

- Local Research Ethics Co-ordinator = LREC
- School level group (if your School uses a different review process, please follow your School guidance)
- University Research Ethics Sub-Committee = URESC

Category	Student applicants	Staff applicants
Risk Category 1	If your study has been provisionally classified as Risk Category 1, your Research Supervisor (or Director of Studies) can normally give approval for the project. You must complete this form and submit it to your Research Supervisor for consideration. A copy of the signed form if approved must be given or emailed to the LREC and, where relevant, the Research Module Leader, for information.	If your study has been classified as Risk Category 1, you do not need ethical approval for the project. You must complete the remainder of this form so that your research project is registered with the University. Please submit this form to your LREC.
Risk Category 2	If your study has been provisionally classified as Risk Category 2, your Supervisor (or Director of Studies) can recommend approval for your study by the LREC. You must complete this application form and also the separate <u>Stage 2 - Research Ethics Approval form</u>. Once you have completed the forms please submit both forms and supporting documentation to your Research Supervisor for consideration. Your Supervisor may disagree with your assessment and ask you to make revisions or reject your application. When the Research Supervisor is happy to recommend the application for approval, they will send the forms to the LREC. The LREC will review your project and then decide to approve it, ask for revisions, reject it or pass it on for review by the School level group.	If your study has been provisionally classified as Risk Category 2, your project will be considered for ethical approval by the LREC. You must complete this application form and also the separate <u>Stage 2 - Research Ethics Approval form</u>. Please submit both forms and supporting documentation to your LREC for consideration. The LREC will review your project and then decide to approve it, ask for revisions or pass it on for review by the School level group.
Risk Category 3	Postgraduate Research Students If your study has been provisionally classified as Risk Category 3, your Supervisor or Director of Studies can recommend approval for your study by the LREC. You must complete this application form and also the separate <u>Stage 2 - Research Ethics Approval form</u> and submit both forms to your Director of Studies. If your Director of Studies recommends approval of your project they will refer it to the LREC who will review your project and decide whether to grant ethical approval, request revisions, reject the application or refer it to the School level group for review. Undergraduate and Taught Postgraduate Students If your study has been provisionally classified as Risk Category 3, you should consult with your Research Supervisor immediately as it is unlikely you will be able to proceed and you should negotiate a project that is of lower risk. However, if you have already discussed the project with your Supervisor and they have agreed that a case for approval is warranted, proceed in line with the details above for Research Students.	If your study has been provisionally classified as Risk Category 3, your project will be considered for ethical approval by an appropriate LREC. You must complete this application form and also the separate <u>Stage 2 - Research Ethics Approval form</u> and submit both forms with supporting documentation to your LREC. The LREC will review your project and then decide to approve it, ask for revisions or pass it on for review by the School level group.
Q23	If question 23 has been answered 'yes', your application will be reviewed by the Chair of the University Research Ethics Sub-committee. The answer does not affect the Risk Category.	

STAGE 2 - RESEARCH ETHICS APPROVAL FORM (December 2016)



STAGE 2 - RESEARCH ETHICS APPROVAL FORM

All research carried out by students and staff at the University must receive ethical approval before any data collection commences.

Notes

- Applicants complete the Risk Checklist and [Stage 1 - Research Ethics Approval Form](#) prior to completing this [Stage 2 - Research Ethics Approval Form](#). Following completion of the Risk Checklist and [Stage 1 - Research Ethics Approval Form](#), if your research study was provisionally classified as Risk Category 2 or 3, you need to complete this form.
- Full details of the project are to be provided in this Stage 2. Where a question in the Risk Checklist was answered YES, please ensure that specific details are included in the appropriate box below.
- If a question does not apply to your project, insert 'Not applicable' or N/A.
- Help is provided for each question. Further help can be found in the Research Ethics Procedures document.
- You navigate through the form by using the tab keys. If you prefer to complete a normal Word document, you can unlock the form by selecting the 'Restrict Editing' button on the Developer tab, then click on 'Stop Protection'. The boxes should expand to allow space for your text.
- Spellchecking is not available in Word forms, so you may find it helpful to prepare your responses in a Word document and then copy these to this form.
- Ensure the form is completed in sufficient detail to allow the reviewer to judge the ethical issues raised by the study. Remember that the reviewer will be considering the following questions when reviewing your application in order to be able to give ethical approval:
 - is it ethical to conduct the research project and is the proposed method of investigation appropriate, thorough and ethical?
 - does the research project meet the requirements of the relevant Research Ethics Principles (Research Ethics Policy A2.4)?

TO BE COMPLETED FOR PROJECTS IN RISK CATEGORY 2 AND 3	
Your name	Aayam Regmi
Project title	Comparative Evaluation of Vision-Based Finger Tracking for Piano Interaction

1	Project Overview
Please give a brief overview of your study, including a summary of your aims and objectives. Help: Describe the purpose of the research and what question(s) the project should answer. To evaluate robustness, accuracy and real-time performance of two or more open-source vision-based finger tracking systems during interaction with a real piano acting as a ground truth. Considering lighting conditions, finger and hand occlusion challenges, and user diversity (hand size and skin tone) the aim is to determine whether these variables affect the consistency, performance and accuracy across different pretrained models in a fast movement environment like a piano.	

STAGE 2 - RESEARCH ETHICS APPROVAL FORM (December 2016)

2	Methodology
Please give a description of your methodology, including any data collection and analysis methods. Help: Give an outline of your study here. If the project is complex, you can also submit your research proposal/protocol (no more than 2-3 A4 sides) if this would help the reviewer's understanding of the project. Include details of your (or your Research Supervisor's) appropriate skills and qualifications to carry out this research.	
Using RAD methodology, after developing the system of data collection pipeline which is a piano recorded by a phone on a tripod, participants involved will play the piano in a certain pre-defined pattern which is friendly to all non-piano players. The system will only record short videos of their hands from the top view and determine the palm-to-palm size of the hands, their skin tone according to Fitzpatrick scale to determine any model biasness when it comes to diversity and occlusion cases where one finger overlap another.	

3	Main Ethical Considerations
Please give a brief description of the main ethical considerations involved in the study. Help: All research projects will have ethical issues, and you will be asked later in the process on recruitment, voluntary participation and the right to withdraw, but highlight here the main ethical considerations for your study (which may concern, e.g., the type of participants, the sensitive nature of the study, the data collection process, a lone researcher carrying out research off-campus, security-sensitive research) and advise how you will address the main issues. If the project is funded, give details here, and whether there are any potential conflicts of interest involved in the study.	
• After being clearly informed about the study's purpose and data collection procedures participants will provide informed consent. • Participation will be entirely voluntary, and participants may withdraw at any time before data anonymisation without penalty. • Only hand regions will be recorded to protect privacy, no identifying features. • All data will be anonymised using participant ID codes instead of names or personal identifiers. • The study does not involve sensitive personal data, vulnerable populations, deception, or invasive procedures. • Collected data (video, landmarks, and MIDI logs) will be stored securely on password-protected and encrypted systems accessible only to the researcher and supervisor. • The tasks involve simple piano key presses and pose no physical or psychological risk beyond everyday activity. • The project is not externally funded and has no conflicts of interest.	

4	Human Participants
----------	---------------------------

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If your study includes Human Participants (or their data), please give a description of who will be included.

Help:

- Please note this should include sample size/number of participants, whether the project will focus on any particular groups/individuals, if it will include any at risk or vulnerable participants, participants aged 16 years or under, etc. Please also specify your rationale for including / excluding groups of participants.
- If the research involves secondary data not in the public domain, give details in this section. The research will only involve video data of the participants hands
 - Relatives
 - Friends
 - Classmates
 - Teachers

5 Recruitment, Voluntary Participation, Consent and Right to Withdraw

If your study includes Human Participants, please give a brief description of the recruitment process, how you will ensure voluntary participation, if (and how) informed consent will be obtained prior to participants taking part in the study, and the right of withdrawal from the research process.

Help:

- This should include clear information on how participants will be identified, approached and recruited; whether the study will include any covert research or deliberate deception; whether help is required from a third party/ gatekeeper to access participants; what information you will give participants, etc.
 - If expenses or any incentives are to be offered to participants, give full details.
 - If your research involves students, colleagues and/or other employees then you must specify the rationale for this and how you will address issues of coercion or feelings of obligation.
 - Regarding withdrawal from the study, discuss the different stages/dates a participant could withdraw or withdraw their data, and how they could do this.
-
- The human participants will have to sign a consent form that will ask for their permission to record their hands.
 - All participants will have the option to withdraw and remove their data even after signing a consent form.

6 Risks and Benefits

Please give a brief description of how, when and where the research will take place and whether there are any risks and/or benefits involved.

Help:

- This should include information on what participants will be required to do, the rationale for this and the level of risk involved.
 - When considering risks, please refer to risks to the participants (e.g., for research in sensitive areas), the researcher, any other parties to the research; and also any health and safety issues for anyone involved (e.g., for lone researchers carrying out fieldwork).
 - If participants will be exposed to ionising radiation, separate approval documentation must be submitted with this application.
-
- The research will take place in a controlled indoor setting (e.g., lab or quiet room) using a fixed webcam positioned above a digital piano.

STAGE 2 - RESEARCH ETHICS APPROVAL FORM (December 2016)

- Data collection sessions will last approximately 10–15 minutes per participant during scheduled project periods.
- Participants will perform simple predefined piano key patterns multiple times under different lighting conditions to evaluate hand-tracking robustness.
- The task involves light key presses only and presents minimal physical or psychological risk beyond everyday activity.
- Only hand regions will be recorded, and all data will be anonymised using participant ID codes.
- Data will be stored securely on password-protected systems accessible only to the researcher and supervisor.
- There are no anticipated health and safety risks to participants, the researcher, or other parties.
- The study may benefit research in hand-tracking robustness, human–computer interaction, and music technology applications.

7	Personal Data, Anonymity and Confidentiality
Please specify what type of information/data will be collected/analysed and the source(s). In addition, specify if and how you will ensure the anonymity of participants and keep information confidential. Help: This should include information on whether you are collecting new information/data or using that that is already in the public domain; whether the data you are using includes personal details; how the data will be processed and stored; who will have access to it; how and when it will be destroyed; the Data Protection requirements for any sensitive personal data, etc. In addition, include whether there may be any requirements for disclosure of information to other parties due to professional practice or legal reasons. If there are limits to confidentiality, explain clearly how the participants would be advised about these limits and possible outcomes.	
• The raw video data will not be published in any platform even GitHub. • After the conclusion and submission of the research all the video data will be removed from the system as the main research data taken out of the video are only numbers in a tabularized file (CSV)	

8	Reporting and Dissemination
Please give details of the planned dissemination and specify if the findings from the research will be published and whether any permission is required for this. Help: This should include information on the methods of dissemination (e.g., dissertation/thesis) and/or what will be published and where (research papers, conference presentations). Specify if any permission is needed (e.g., from participants, clients, gatekeepers, etc.) prior to publication, and whether there are any potential issues relating to Intellectual Property Rights when creating or using materials.	
• The findings will be primarily disseminated through the final year dissertation submitted to the university for academic assessment. • Results may also be presented in academic presentations, student research showcases, or potentially submitted to relevant conferences or journals in machine learning or human–computer interaction. • Only anonymised and aggregated results (e.g., model performance metrics, error rates, statistical analysis) will be reported in any publication. • No identifiable participant information, images of faces, or personal data will be published. • Participants will provide informed consent that includes permission for anonymised data to be used in academic publications or presentations.	

STAGE 2 - RESEARCH ETHICS APPROVAL FORM (December 2016)

- There are no anticipated Intellectual Property conflicts, as the dataset will be newly generated for academic research purposes.
- Any third-party software or models used (e.g., open-source hand-tracking frameworks) will be properly cited in accordance with their licensing requirements.

9	Location of research
Will the research take place outside of the country where you are enrolled as a student, or for staff, outside of the UK?	
YES ☒ NO ☐ If yes, give details below.	
<u>Help:</u> If yes, please specify where the research will take place and what will be involved. Research must comply with the laws of the country where it is taking place and also comply with local Data Protection and Intellectual Property legislation: you must confirm that your research is compliant with local requirements and how you have ascertained this. Advise if the project requires ethical approval in-country and how this has been ascertained. If approval is required, a copy of this should be included in the application or details of the process of how it will be obtained. Please make reference to insurance and indemnity cover for the project where relevant.	
The British College, Kathmandu The research complies to all law and ethical considerations of Nepal	

10	Collaborative Projects
Is the research is a collaborative project (i.e., it involves more than one institution)?	
YES ☐ NO ☒ If yes, give details below.	
<u>Help:</u> If yes, please specify the other institutions involved and if ethical approval needs to be / has been given by them. Please also specify what procedures have been put in place to ensure ethical compliance from all partners.	

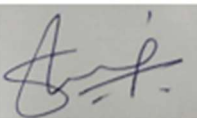
11	Any other permission or external ethical approval required to undertake the project
Please specify if the project requires any other ethical approval or permissions not mentioned previously in this application and how and when these will be obtained.	
<u>Help:</u>	

STAGE 2 - RESEARCH ETHICS APPROVAL FORM (December 2016)

- Other permissions: ethical approval does not give the right of access to the University's students, staff or the use of university premises to carry out research, and you may need to contact an appropriate University gatekeeper for agreement to approach potential participants or for the use of premises, so please give details.
- Gatekeepers: permission of a gatekeeper for initial access to participants may be required or to carry out data collection on their premises.
- If your project requires approval from an external ethics committee, this should normally be obtained prior to submitting this application.
- If a Disclosure and Barring Service check is required due to the specific participant group, give details.
- Regarding insurance and indemnity cover, some projects will require individual confirmation of cover. See the Research Ethics Procedures document for more details.

STAGE 2 - RESEARCH ETHICS APPROVAL FORM (December 2016)

FOR PROJECTS INVOLVING RISK CATEGORY 2 AND 3: DECLARATION AND SIGNATURE/S			
APPLICANT (STUDENT/STAFF MEMBER/RESEARCHER)			
I confirm that I will undertake this project as detailed in stage one and stage two of the application. I understand that I must abide by the terms of this approval and that I may not make any substantial amendments to the project without further approval. I understand that research with human participants or their data must not commence without ethical approval.			
I have read an appropriate professional or learned society code of ethical practice:		Yes ☒ N/A ☐	
Where applicable, give the name of the professional or learned society:		Aayam Regmi	
Signed		Date	3 rd March, 2026

RESEARCH SUPERVISOR/DIRECTOR OF STUDIES RECOMMENDATION FOR STUDENT PROJECTS				
I confirm that I have read stage one and stage two of the application. The project is viable and the student has appropriate skills to undertake the project. Where applicable, the Participant Information Sheet and recruitment procedures for obtaining informed consent are appropriate and the ethical issues arising from the project have been addressed in the application. I understand that research with human participants must not commence without ethical approval. I recommend this project for approval.				
Name	Mr. Saroj Sharma	Signed		Date
				3 rd March, 2026

Local Research Ethics Co-ordinators

Please complete EITHER A (giving ethical approval for the project) OR B (recommending the project to the School level group for approval)

A	LOCAL RESEARCH ETHICS CO-ORDINATOR APPROVAL For projects approved by the Local Research Ethics Co-ordinator			
I confirm ethical approval for this project				
LREC Name		Signed		Date

OR

B	LOCAL RESEARCH ETHICS CO-ORDINATOR'S RECOMMENDATION FOR SCHOOL APPROVAL For projects that require School level approval			
I recommend this project for consideration at school level. It cannot be approved at local level due to the following reason(s)				

STAGE 2 - RESEARCH ETHICS APPROVAL FORM (December 2016)

LREC Name		Signed		Date	
--------------	--	--------	--	------	--

School level group

For projects approved at School level please complete the box below.

PROJECTS APPROVED BY THE SCHOOL LEVEL GROUP					
I confirm that this project was considered by the School level group and has received ethical approval					
Group Lead		Signed		Date	

OR

University Research Ethics Sub-Committee

For projects approved by URESC please complete the box below.

Projects involving security-sensitive research do not need supervisor/LREC approval prior to being considered by the Chair of URESC.

PROJECTS APPROVED BY THE RESEARCH ETHICS SUB-COMMITTEE					
I confirm that this project was considered by the Research Ethics Sub-committee and has received ethical approval					
Chair		Signed		Date	

This form will be retained for the purposes of quality assurance of compliance and audit for THREE years

STAGE 2 - RESEARCH ETHICS APPROVAL FORM (December 2016)

SUPPORTING DOCUMENTATION: what to submit with the application

For projects involving human participants, you must submit, where appropriate, the Participant Information Sheet/s and consent form/s. You must also submit every communication a participant will see or receive. Failure to do so will cause delays to the application.

Below is a checklist reminder of what could be submitted, depending on the research project. Please tick the appropriate boxes for each attachment or give details of the document at the end of the checklist.

SUBMISSION CHECKLIST	Tick box
RISK CHECKLIST AND STAGE 1 – RESEARCH ETHICS APPROVAL FORM	☒
STAGE 2 – RESEARCH ETHICS APPROVAL FORM	☒
Participant Information Sheet(s)	☐
Consent Form(s)	☒
Assent Form (usually for children participants)	☐
Recruitment documents <i>eg. posters, flyers, advertisements, email invitations, letters, web pages if online research</i>	☐
Measures to be used <i>eg. questionnaires, surveys, interview schedules, psychological tests</i>	☐
Screening questionnaire	☐
Letters/communications to and from gatekeepers/third parties	☐
Evidence of any other approvals or permissions <i>eg. NHS research ethics approval, in-country approval</i>	☐
Research proposal/protocol (no more than 2-3 A4 pages) <i>It is not a requirement that this is included, however, if this would help the understanding of a complex project by the reviewer(s), please include</i>	☐
Risk assessment form <i>Some projects may require a risk assessment form: see the Procedures document for details (eg. projects involving a physical intervention, collecting data off-campus)</i>	☐
Approval documentation for projects involving ionising radiation	☐
Confirmation of insurance and indemnity cover where relevant <i>Some projects need to be referred to the Insurance & Risk Officer: see the Procedures document</i>	☐
Security-sensitive research form	☐
Other: give details here:	☐
	☐

SUBMITTING YOUR FORMS

- Students: email the typed forms (stage one and stage two) and supporting documentation to your Research Supervisor or Director of Studies.
- Staff: email the typed forms (stage one and stage two) and supporting documentation to your Local Research Ethics Co-ordinator.
- Security-sensitive research: the stage one form (and stage two form if applicable) should be submitted directly to the URES Chair, Professor Karl Spracklen, k.spracklen@leedsbeckett.ac.uk and include the Security-sensitive research form, available from the Research Ethics web page.

D. Participant Consent Form

PARTICIPANT CONSENT FORM

Project Title: Comparative Evaluation of Vision-Based Finger Tracking for Piano Interaction

Researcher: Aayam Regmi — BSc Student, Leeds Beckett University

Student Number: 77466864

Please read each statement carefully and initial the box if you agree.

I confirm that I have read and understood the Participant Information Sheet for the above study and have had the opportunity to ask questions.	✓
I understand that my participation is voluntary and that I am free to withdraw at any time up until my data is anonymised and included in the final report, without giving any reason and without consequence.	✓
I consent to the session being recorded (video footage) and to MIDI performance data being captured during the study.	✓
I understand that recordings and session data will be stored securely and used solely for the purposes of this research project. Video recordings will not be shared publicly.	✓
I understand that no data allowing identification of me as an individual will be published. I will be referred to only by an anonymous participant code (e.g. P001).	✓
I consent to my anonymised data (including Fitzpatrick skin type, hand size, lighting conditions, and performance metrics) being used in the final research report and any associated publications.	✓
I understand that the information I provide will be published as part of a research report. Confidentiality and anonymity will be maintained and it will not be possible to identify me from any publications.	✓

Participant:

Name : _____

Signature: _____

Date: 10th March, 2026

Researcher:

Name (print): Aayam Regmi

Signature:

Date: 28th February, 2026

E. Source Code Snippets

config.py

Project-wide configuration constants for camera, model and threshold parameters.

```
CAMERA_INDEX = 0
CAP_WIDTH = 1280
CAP_HEIGHT = 720
CAP_FPS = 60
MODEL_COMPLEXITY = 1
MAX_NUM_HANDS = 2
MIN_DETECTION_CONFIDENCE = 0.6
MIN_TRACKING_CONFIDENCE = 0.6
LUX_DIM_THRESHOLD = 100
LUX_BRIGHT_THRESHOLD = 500
```

Centralises all tunable values so changes do not require editing multiple scripts.

camera_setup.py

Initialises the connected USB camera with the configured resolution and frame rate.

```
def init_camera(config):
    cap = cv2.VideoCapture(config.CAMERA_INDEX, CAP_API)
    cap.set(cv2.CAP_PROP_FRAME_WIDTH, config.CAP_WIDTH)
    cap.set(cv2.CAP_PROP_FRAME_HEIGHT, config.CAP_HEIGHT)
    cap.set(cv2.CAP_PROP_FPS, config.CAP_FPS)
    cap.set(cv2.CAP_PROP_BUFFERSIZE, config.CAP_BUFFERSIZE)
    if not cap.isOpened():
        sys.exit("Error: could not open camera.")
    return cap
```

Opens the camera via DirectShow, applies the requested capture parameters, and returns the handle for the recording loop.

lux_calculator.py

Maps a manually entered lux value to a lighting category label used in analysis.

```
def lux_to_label(lux: float) -> str:
    if lux >= config.LUX_BRIGHT_THRESHOLD:
        return "Bright"
    if lux >= config.LUX_DIM_THRESHOLD:
        return "Indoor"
    return "Dim"
```

Converts the raw lux measurement into one of three category labels (Dim, Indoor, Bright) using the thresholds defined in config.py.

fitzpatrick_detector.py

Estimates Fitzpatrick skin type from the palm region of a frame using the Individual Typology Angle.

```
_PALM_IDS = [0, 1, 5, 9, 13, 17]

def _ita_to_fitzpatrick(ita_angle: float):
    for threshold, ftype, label in _ITA_THRESHOLDS:
        if ita_angle > threshold:
            return ftype, label
    return 6, "VI - Very Dark"
```

Maps an ITA angle to one of six Fitzpatrick categories using fixed thresholds from Chardon et al. (1991).

```
lab = cv2.cvtColor(frame_bgr, cv2.COLOR_BGR2Lab)
lab_pixels = lab[mask == 255]
L_star = float(np.mean(lab_pixels[:, 0])) * (100.0 / 255.0)
b_star = float(np.mean(lab_pixels[:, 2])) - 128.0
ita = np.degrees(np.arctan((L_star - 50.0) / b_star))
```

Samples Lab colour values from the masked palm region and computes the ITA angle from the mean L* and b* components.

midi_recorder.py

Background MIDI listener that captures note events with sub-frame timestamps.

```
def start(self, t0: float) -> None:
    self._t0 = t0
    self._stop_evt.clear()
    self._port = mido.open_input(self.port_name)
    self._thread = threading.Thread(
        target=self._listen, daemon=True, name="MidiListener")
    self._thread.start()
```

Opens the MIDI input port and starts a background daemon thread anchored to the shared time origin.

```
def _listen(self) -> None:
    while not self._stop_evt.is_set():
        for msg in self._port.iter_pending():
            t_now = time.perf_counter() - self._t0
            with self._lock:
                self._events.append((t_now, msg))
            time.sleep(0.001)
```

Polls the MIDI port every millisecond and stores each event with a perf_counter timestamp relative to recording start.

model_manager.py

Abstract pose-model wrapper providing a uniform interface for MediaPipe and OpenPose.

```
class PoseModel(abc.ABC):
    @abc.abstractmethod
    def load(self) -> None: ...
    @abc.abstractmethod
    def infer(self, bgr: np.ndarray) -> object: ...
    @abc.abstractmethod
    def draw(self, frame: np.ndarray, result: object) -> np.ndarray: ...
    @abc.abstractmethod
    def close(self) -> None: ...
```

Defines a common abstract interface so the recording pipeline can swap models without changing call sites.

```
class MediaPipeHandsModel(PoseModel):
    def load(self) -> None:
        import mediapipe as mp
        self._hands = mp.solutions.hands.Hands(
            static_image_mode=False,
            max_num_hands=self._cfg.MAX_NUM_HANDS,
            model_complexity=self._cfg.MODEL_COMPLEXITY,
            min_detection_confidence=self._cfg.MIN_DETECTION_CONFIDENCE,
            min_tracking_confidence=self._cfg.MIN_TRACKING_CONFIDENCE)
```

Lazily initialises MediaPipe Hands with the parameters from config.py on first use.

stats_collector.py

Tracks running per-second statistics during live recording (FPS, latency, detection counts).

```
class StatsCollector:
    def __init__(self):
        self.frames = 0
        self.detections = 0
        self.t_start = time.perf_counter()
    def add_frame(self, detected: bool):
        self.frames += 1
        if detected: self.detections += 1
    def fps(self) -> float:
        return self.frames / max(time.perf_counter() - self.t_start, 1e-6)
```

Counts frames and successful detections since the start, exposing the running FPS for the live overlay.

key_calibration.py

Interactive interface for defining the per-key polygon mask aligned to the keyboard.

```
def build_key_layout(start_midi, num_white, ox, oy, wkw, wkh) -> list:
    keys = []; white = 0
    for midi in range(start_midi, 128):
        semi = midi % 12
        if semi in _WHITE_SEMIS:
            x = ox + white * wkw
            keys.append({'midi_note': midi, 'key_type': 'white',
                        'rect': (x, oy, wkw, wkh),
                        'center': (x + wkw // 2, oy + int(wkh * 0.82))})
            white += 1
        else:
            cx = ox + white * wkw
            keys.append({'midi_note': midi, 'key_type': 'black',
                        'rect': (cx - bkw // 2, oy, bkw, bkh),
                        'center': (cx, oy + bkh // 2)})
    return keys
```

Generates pixel rectangles and center points for each white and black key from a single calibration anchor.

```
M = cv2.getPerspectiveTransform(src_quad, dst_quad)
warped = cv2.perspectiveTransform(np.array([key_polygons], dtype=np.float32), M)[0]
```

Applies a homography transform to warp the planar key grid onto the camera's oblique projection of the keyboard.

record.py

Main recording entry point that captures synchronized video and MIDI for one session.

```
def _start_recording(frame, pid: int, fps: float) -> tuple:
    out_dir = _RAW_DIR / f"p{pid:03d}"
    out_dir.mkdir(parents=True, exist_ok=True)
    h, w = frame.shape[:2]
    fourcc = cv2.VideoWriter_fourcc(*"mp4v")
    writer = cv2.VideoWriter(str(out_dir/"video.mp4"), fourcc, fps, (w, h))
    return out_dir, writer
```

Creates the per-participant output directory and opens the video writer at the camera's reported FPS.

```
t0 = time.perf_counter()
midi_rec.start(t0)
frame_log = []
while recording:
    ok, frame = cap.read()
    if not ok: break
    t_frame = time.perf_counter() - t0
    writer.write(frame)
    frame_log.append((t_frame, frame_idx))
    frame_idx += 1
```

Anchors video and MIDI to a shared perf_counter origin so every frame timestamp is comparable to every MIDI event.

```
def _save_session_metadata(out_dir, pid, lux_value, lux_label,
                          fitz_type, fitz_label, hand_size_cm, hand_size_label):
    with (out_dir/"session.json").open("w") as f:
        json.dump({"pid": pid, "lux": lux_value, "lux_label": lux_label,
                  "fitzpatrick": fitz_type, "fitzpatrick_label": fitz_label,
                  "hand_size_cm": hand_size_cm,
                  "hand_size_label": hand_size_label}, f, indent=2)
```

Persists the participant's environmental and demographic variables into a session JSON for later cross-analysis.

[live_preview.py](#)

Real-time overlay window showing landmarks, key polygons, FPS and per-finger results.

```
def _resolve_hand_label(hand_lms, multi_handedness, i: int, fw: int) -> str:
    wrist_x = hand_lms.landmark[0].x * fw
    thumb_mcp_x = hand_lms.landmark[2].x * fw
    geo_label = 'L' if thumb_mcp_x > wrist_x else 'R'
    if multi_handedness and i < len(multi_handedness):
        mp_raw = multi_handedness[i].classification[0].label
        mp_label = 'L' if mp_raw == "Left" else 'R'
        return mp_label if mp_label == geo_label else geo_label
    return geo_label
```

Combines MediaPipe's handedness label with thumb-wrist geometry to produce a mirror-safe physical hand label.

```
for slot, (x, y) in fingertips.items():
    cv2.circle(display, (x, y), 5, (0, 255, 0), -1)
    cv2.putText(display, _FINGER_NAMES[slot], (x+6, y-6),
                cv2.FONT_HERSHEY_SIMPLEX, 0.4, (255, 255, 255), 1)
```

Draws each detected fingertip with a coloured marker and finger name on the live preview frame.

recalibrate.py

Standalone tool to re-warp the polygon mask of a previously recorded session.

```
def _load_frame_times(frames_csv: Path, video_fps: float, total_frames: int) ->
list[float]:
    times = []
    with frames_csv.open() as f:
        reader = csv.reader(f)
        next(reader) # header
        for row in reader:
            times.append(float(row[1]))
    return times
```

Reads per-frame timestamps from the frames CSV so the editor can scrub the video while preserving the original sync.

analyse.py

Offline pipeline that runs both models against each recorded session and computes MPJPE.

```
def _nearest_frame_idx(event_time: float, frame_times: list) -> int:
    ts_list = [t for t, _ in frame_times]
    i = bisect.bisect_left(ts_list, event_time)
    i = min(max(i, 0), len(ts_list) - 1)
    if i > 0 and abs(ts_list[i-1] - event_time) < abs(ts_list[i] - event_time):
        i -= 1
    return frame_times[i][1]
```

Performs a binary-search lookup to find the video frame nearest in time to a given MIDI event.

```
def _hand_label_from_mp(hand_lms, multi_handedness, i: int, fw: int) -> str:
    wrist_x = hand_lms.landmark[0].x * fw
    thumb_mcp_x = hand_lms.landmark[2].x * fw
    geo_label = 'L' if thumb_mcp_x > wrist_x else 'R'
    if multi_handedness and i < len(multi_handedness):
        mp_raw = multi_handedness[i].classification[0].label
        mp_label = 'L' if mp_raw == "Left" else 'R'
        return mp_label if mp_label == geo_label else geo_label
    return geo_label
```

Validates MediaPipe's handedness label against thumb-wrist geometry to handle mirrored vs non-mirrored cameras.

```

for tip in tips:
    slot, x, y, hand = tip
    if cv2.pointPolygonTest(polygon, (x, y), False) >= 0:
        err = abs(x - key_center_x)
        if err < best_err:
            best_err, best_tip = err, tip
if best_tip is not None:
    matched.append((slot, hand, best_err))

```

For each MIDI note, selects the closest fingertip lying inside the triggered key's polygon as the matched event.

[plot_results.py](#)

Generates the figures used in the Results chapter from the per-session JSON outputs.

```

def plot_model_comparison(records, out_dir):
    by_model = defaultdict(list)
    for r in records:
        if r.get("mjpe_px") is not None:
            by_model[r["model"]].append(r["mjpe_px"])
    means = [np.mean(by_model[m]) for m in models]
    sds = [np.std(by_model[m]) for m in models]
    fig, ax = plt.subplots(figsize=(6, 5))
    ax.bar(np.arange(len(models)), means, yerr=sds, capsize=6)
    _save(fig, out_dir, "01_model_comparison_mjpe.png")

```

Plots the overall MediaPipe vs OpenPose MPJPE comparison with mean $\pm$ SD error bars using matplotlib.

```

def plot_per_finger_mjpe(records, out_dir):
    fig, axes = plt.subplots(1, 3, figsize=(16, 5))
    for finger in range(5):
        axes[0].bar(finger, _finger_vals(records, 'L', finger))
        axes[1].bar(finger, _finger_vals(records, 'R', finger))
        axes[2].bar(finger, _finger_vals_combined(op_records, finger))
    _save(fig, out_dir, "02_per_finger_mjpe.png")

```

Plots per-finger MPJPE in three side-by-side panels: MediaPipe Left, MediaPipe Right, and OpenPose combined.

[export_table_data.py](#)

Collates exact numerical values from every plot into a single JSON for use in tables.

```
def chart_03(records):
    out = {}
    for model in ["mediapipe", "openpose"]:
        rows = [r for r in records if r["model"] == model]
        m = sum(r["notes_matched"] for r in rows)
        f = sum(r["notes_detection_fail"] for r in rows)
        x = sum(r["notes_missed"] for r in rows)
        total = m + f + x
        out[model] = {"matched_count": m, "detection_fail_count": f,
                     "missed_count": x, "total_count": total,
                     "matched_pct": round(m/total*100, 2)}
    return out
```

Aggregates the detection breakdown counts and percentages across all sessions for both models.

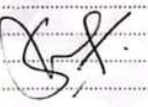
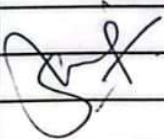
[data_summary.py](#)

Quick command-line summary of total participants, frames and MIDI events recorded.

```
def main():
    sessions = sorted(p for p in PROC_DIR.glob("p*_mediapipe_results.json"))
    total_frames = total_midi = total_notes = 0
    for path in sessions:
        with path.open() as f:
            d = json.load(f)
            total_notes += d["notes_matched"] + d["notes_detection_fail"] + d["notes_missed"]
    print(f"Sessions: {len(sessions)} Notes: {total_notes}")
```

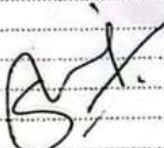
Walks the processed JSON folder and prints a one-line dataset summary covering session count and total events.

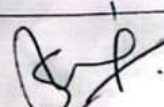
F. Supervisor Meeting Records

School of Computing, Creative Technologies and Engineering 2024/25	
Level 6 Production Project	
MEETING RECORD SHEET:	Meeting Number: <u>1</u>
Student: <u>Aayam Regmi</u>	Student I.D.: <u>77466864</u>
Date of Meeting: <u>23rd Feb, 2026</u>	Supervisor: <u>Mr Suroj Sharma</u>
Actions agreed at previous meeting (completed or comment):	
1	☐
2	☐
3	☐
4	☐
5	☐
6	☐
Comments of student (if any):	
<u>Meeting</u>  	
ABOVE here - student to complete before Meeting with supervisor. BELOW here - complete at the Meeting.	
Next meeting (date/time): <u>2nd March, 2026</u>	
Agreed Actions to complete before next meeting:	
1	<u>Review Similar research papers of AR apps</u>
2	<u>See how MediaPipe, OpenPose and Movenet works</u>
3	
4	
5	
6	

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School of Computing, Creative Technologies and Engineering 2025/26	
Level 6 Production Project	
MEETING RECORD SHEET:	Meeting Number: 2
Student: Aayam Regmi	Student I.D.: 17466864
Date of Meeting: 2 nd March, 2026	Supervisor: Mr. Saroj Sharma
Actions agreed at previous meeting (completed or comment):	
1	☐
2	☐
3	☐
4	☐
5	☐
6	☐
Comments of student (if any):	
project brief <i>[Signature]</i>	
ABOVE here - student to complete before Meeting with supervisor. BELOW here - complete at the Meeting.	
Next meeting (date/time): 9 th Feb March, 2026	
Agreed Actions to complete before next meeting:	
1	do a research on the existing system
2	
3	<i>[Signature]</i>
4	
5	

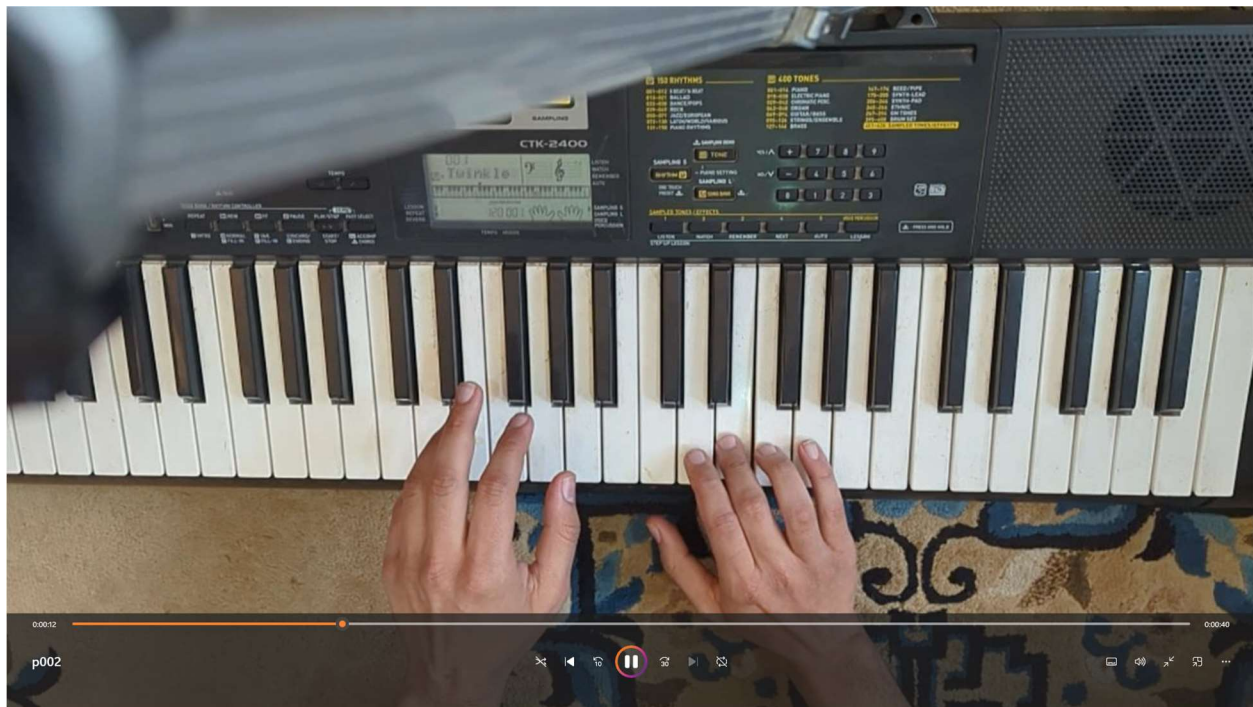
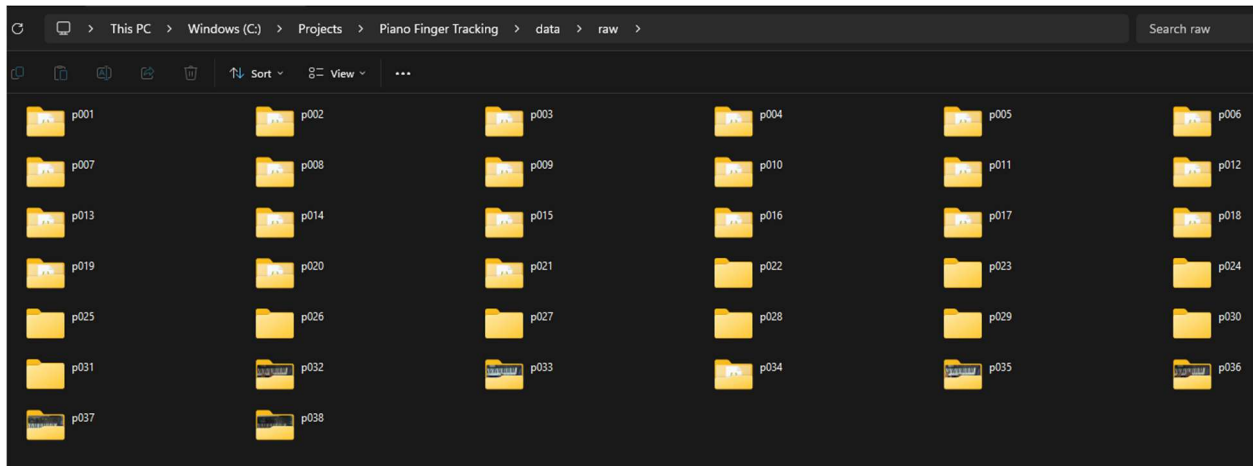
School of Computing, Creative Technologies and Engineering 2025/26	
Level 6 Production Project	
MEETING RECORD SHEET:	Meeting Number: 3
Student: Aayam Regmi	Student I.D.: 77466864
Date of Meeting: 9 th March, 2026	Supervisor: Mr. Sazaj Sharma
Actions agreed at previous meeting (completed or comment):	
1	Review existing research on vision models ☒
2	☐
3	☐
4	☐
5	☐
6	☐
Comments of student (if any):	
	
ABOVE here - student to complete before Meeting with supervisor. BELOW here - complete at the Meeting.	
Next meeting (date/time): 6 th April, 2026	
Agreed Actions to complete before next meeting:	
1	Develop data collection pipeline
2	Develop data characterization variables
3	Make UI for pipeline
4	Collect raw video data

School of Computing, Creative Technologies and Engineering 2025/26	
Level 6 Production Project	
MEETING RECORD SHEET:	Meeting Number: 4
Student: Aayam Regmi	Student I.D.: 77466864
Date of Meeting: 8 th April, 2026	Supervisor: Mr Sargj Sharma
Actions agreed at previous meeting (completed or comment):	
1	Develop data collection pipeline ☒
2	Develop data characterization variables ☒
3	Make UI for pipeline ☒
4	Collect raw video data (20 collected) ☐
5	☐
6	☐
Comments of student (if any):	
	
ABOVE here - student to complete before Meeting with supervisor. BELOW here - complete at the Meeting.	
Next meeting (date/time): 13 th April, 2026	
Agreed Actions to complete before next meeting:	
1	Complete data collection
2	Produce charts showing MIDI sync data
3	
4	
5	

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23

G. Data Collection Source



P002 recording data